

Methods

Animals

Animals: We used 7-9 weeks old (n = 10 mice) female and male C57BL/6J mice. Six animals were pooled in the first-session group and four in the expert-session group. Blinding is not relevant to this study since all behavioral and neuronal activity analyses were performed with automated scripts without experimenter intervention or selection, and curation of spike-sorting results was done without knowledge of responsiveness to stimuli of each single-unit cluster. Littermates were randomly assigned to first-session or expert-session groups. No statistical methods were used to predetermine sample size. Mice were group-housed before head bar implantation and housed singly thereafter. The experimental timeline is shown in Extended data Fig. 1a.

Water Restriction Protocol: Water restriction started after mice recovered from surgery (at least three days post-surgery, Extended data Fig. 1a). Mice were individually housed in cages containing play tunnels, a wheel, and nesting material, in a reverse light cycle (12O:12L) housing room. The light in the accommodation room had an intensity of 100 lux at the height of the boxes, with white LEDs. The habituation, training and neuronal activity recording sessions were carried out during the dark phase. In the experimental device where the animals were placed, there was no illumination besides the light from the virtual reality screen. Relative humidity was maintained at 40-50%. Dry food was continuously available. One ml of water was dispensed manually into small containers attached to the inner walls of the cages, always at the same time of day. Mice were monitored daily for hydration level, weight, skin health, and locomotion. To control these parameters, a daily record of: movement in the box, cleaning/grooming, posture, tension/relaxation of the skin of the neck, defecation in the housing box was used. Our protocol (IBioBA-CICUAL # 2020-03-NE) was subject to rigorous animal welfare control measures, including daily weight

determinations and activity scores, posture, grooming, intake, and signs of dehydration. In the event of a weight loss below the minimum body weight (70% of the initial weight), the total daily volume of water they receive was increased until the stabilization of the body weight in water restriction.

Behavioral task

Mice were head-fixed and trained to run on a running wheel (a plastic cylinder) (Fig. 1a and Extended data Fig. 1b). A water spout was positioned near the snout of the animal, and licks were detected with an infrared beam and sensor. Wheel rotations as the animal ran were translated into displacements through a linear virtual corridor, which was presented in a screen in front of the animal. Mice were trained in a behavioral task of olfactory discrimination dependent on visual context. Here, visual context refers to the virtual environment, which could be one of two types of corridors. Each corridor consisted of an approaching aisle (83 cm in length), a context zone (33 cm in length) and a reward zone. Both types of corridors were visually distinct only in the context zone. The approaching aisle and the reward zone were similar for both corridors, with black walls displaying white spots. The context zone could be either green (green floor and walls with vertical white and green stripes, and a green column on the right with a black diamond-shaped pattern) or gray (black floor with white diamond-shaped pattern and gray walls with black dots, and a column on the left with horizontal white, black and gray stripes) (Extended data Fig. 1b). When the animals entered the context zone, they received an odor puff of 1-second duration through a pipe directed to the animal nose. The odorant could be of two types: isoamyl acetate or ethyl butyrate. Once animals left the context zone where they received the odorant stimulus, they entered the reward zone, where they could choose to lick (GO response) or not (NO-GO response) to obtain a drop of water reward. There are four possible odor-context combinations, only one of these was rewarded after the GO response in the reward zone (Fig. 1b). No punishment was given for incorrect GO

responses or for licking elsewhere in the corridor in any trial type. After reward delivery or an incorrect GO response the trial was immediately finished and a new trial started by “teletransporting” mice to the start of the approaching aisle.

A specific rewarded odor-context combination (e.g., isoamyl acetate-gray context) was randomly assigned to each mouse and maintained throughout all training sessions. This rewarded combination was labeled $O_R C_R$. The individual odorant and context used in that rewarded combination were labeled O_R and C_R respectively, while the remaining odorant and context were labeled O_U and C_U . The “R” and “U” subscripts stand for “rewarded” and “unrewarded”, but notice this is an abuse of language since specific odor-context combinations were rewarded, not individual odors or contexts (that is, only $O_R C_R$ trials were rewarded while $O_R C_U$ and $O_U C_R$ trials were not).

Passive stimulation: for 4 of the first-session and 4 of the expert animals, at the end of the behavioral task we additionally performed a passive odorant stimulation protocol. The protocol consisted of 20 trials of stimulation with each odorant (trials were randomly interleaved), while the virtual reality monitor was turned off (Fig. 6a). Odors pulses lasted 1 second, and were presented every 40 seconds.

Experimental apparatus: The device consisted of: I. Columns for holding the animal's head (head-fixed rig). Animals are fixed to the rig by clamping a skull-implanted metal bar (headbar) to the rig's columns. II. A running wheel for the animal with a rotary encoder attached to the wheel's axle that records wheel rotations as the animal walks (see *Data acquisition*). The wheel is made of plastic and is covered with black EVA rubber so that the substrate is soft to the touch of the animal and avoids slippage; III. A circuit of valves connected to pipes (olfactometer) that conducts the odor to the animal's nose (see *Olfactometer in Odor Stimuli*). This system is computer-controlled, which allows different odors to be presented with adjustable flow rates and to switch between

odors quickly (<20 ms). In addition, an air exhaust quickly removes the odor after its presentation. Both the olfactometer and the exhaust allow the olfactory stimulus to be presented in a timely manner. Airflow is kept constant and regulated by flowmeters; IV. An inhalation-exhalation cycle measurement system: sniffing behavior was recorded with a mass airflow sensor located externally in close proximity to the animal's left nostril (see *Data Acquisition and Analysis of respiration recordings*). Precise orientation relative to the nostril was manually optimized before each recording to attempt to acquire a full signal, despite any movement of the nose. V. A custom-made lickometer for measuring the animal's licking response and for delivery of water rewards (see *Data Acquisition*). VI. A computer screen placed in front of the animal that displayed the virtual reality environment according to the task protocol. VII. A microcontroller and data acquisition system (Bpod State Machine r1, Sanworks) that records the animal's licking response and its behavioral performance across trials, while controlling in real time the state of the task (activation of odor valves, delivery of reward, sequence of trial types) according to the task protocol. VIII. A data acquisition system (Smartbox, Neuronexus) that acquires and stores behavioral and trial information, along with the electrophysiological signals (see *Data Acquisition*). IX. A CPU running MATLAB (Mathworks) for controlling the BPod and running the Virtual Reality MATLAB Engine, ViRMEn⁵⁷.

Animal training

Behavioral training: After body weight stabilization, usually after five days of water restriction, the experimental device habituation and training session began (Extended data Fig. 1a). During those days, we carried out a daily *handling* or manipulation session in which the animals got used to the manipulation of the experimenter and to remain comfortable in the hands of the experimenter. *Habituation to the experimental device:* Three days before training, the mice were

manipulated daily so that they got used to the environment of the training room and head fixation to the experimental device. *Behavioral task training:* A random sequence of trial types (different odor-context combinations) was presented to the animal in each training session. Individual training sessions lasted 40-60 minutes. In the first training session animals completed 134 +/- 51 trials, while they performed 227 +/- 60 trials on expert sessions. Rewards were 10 microliter water drops.

Odor stimuli

Olfactometer: Odors were delivered using a custom made olfactometer. Charcoal filtered air was routed into two flowmeters (03216-06 and 03216-16, Cole-Parmer, IL). The first one carried a neutral air stream at 0.9 liters per minute (LPM) that was kept constant (base stream). The second one carried a 0.2 LPM stream which was further splitted in two 0.1 LPM arms with an injection valve (SI360T041, NResearch, NJ; “odor bank valve”). When the valve was turned off the first arm was selected and the air stream traversed through a 10 mL empty vial containing (blank stream). When the valve was turned on the second arm was selected and the air stream was channeled to a second injection valve (“odor selection valve”) that routed the air to one of two 10 mL vials, each one containing a 2 mL solution of one of the two odors. The voltage state of this odor selection valve thus controlled which odor would compose an odorant stream. The two arms were combined in a shuttle valve (SH360T041, NResearch, NJ; “isolation valve”) that was synchronously activated with the odor selection valve to isolate both odor paths, avoiding odorant cross-contamination that could result from air reflux into the remaining vial. The odorant stream and the blank stream were then fed to a final shuttle valve that selected between both streams (“injection valve”). The selected 0.1 LPM stream (odorant or blank) was combined with the 0.9 LPM base stream to produce a 1 LPM stream that was routed to the animal nose. The Bpod system was programmed to control valve voltages so that when entering the visual context, the appropriate

odorant stream would be selected and the animal would receive the 1-second-long odor pulse scheduled for that trial. Final valve switching simultaneously rerouted the odorant stream to the animal's nose and the blank stream to a vacuum exhaust, and switched back after 1 second. At any other trial moment, the blank stream would be selected and the animal would receive an air stream that passed through vials containing only mineral oil. Non-selected odorant streams were also directed to the vacuum exhaust. Consistency of shape and arrival times of odor pulses were monitored with a custom-made photoionization detector (PID; PID-A1, Alphasense, Essex, UK) located at the outlet of the stimulation tubing near the animal's nose. Calibration of the olfactometer was routinely performed such that switching of the injection valve produced minimal perturbations of the total air stream to the animal's nose. Tubings were made of Teflon to avoid accumulation of residual odors.

Odorant stimuli: isoamyl acetate or ethyl butyrate (SIGMA-Aldrich). The odors were chosen based on literature showing that these two odors had no innate response in mice⁶. The odors mentioned were prepared in the liquid phase at room temperature and the volatiles of the gaseous phase were supplied to the animals as a stimulus. For all experiments, solutions with 1:200 dilutions were used (odor: mineral oil) for odorant vials. The volatiles present in the gas phase of the vials were administered to the animals through the olfactometer. Odorant and blank vials in the olfactometer contained a liquid phase of 2000 µl of the respective solutions and a gas phase of 10 ml in which vapors were accumulated until reaching equilibrium with their respective vapor pressure. As a source of purified air, an aquarium pump was used whose air was passed through activated carbon and cotton filters. Airflow was constantly controlled by valves driven by electrical controllers. The opening and closing of the air flow passing through each vial was regulated. When a valve was opened, a volume of the gaseous portion of the vial was displaced, mixed with a continuous main

flow of purified air, and finally emptied into a tube with an outlet located 1-2 cm from the mouse's head.

Surgeries

Stereotactic targeting and head bar attachment surgery: Animals were injected intraperitoneally with Ketamine (100mg/kg body weight) and Xylazine (10mg/kg body weight). From the establishment of total anesthesia, controlled through the loss of body reflexes, the operation lasted a maximum of 30 min. It was done on a heating pad to preserve the animal's body heat. The eyes were protected from drying out by applying ophthalmic gels such as Vidisic (active ingredient: polyacrylic acid). The entire procedure was performed under aseptic conditions. Once anesthetized, animals were placed in a stereotaxic. A dose of bupivacaine 0.5% (50µL) was injected under the skin as a local anesthetic, and we waited for 5 minutes before continuing. An incision was made in the skin to expose the skull, disinfecting the area with povidone using sterile swabs. The animal was prepared to place a metal bar that allowed us to attach its head to the experimental device and keep it fixed, and on the other hand to mark the skull in the position where the recording electrode would be inserted afterwards. The scalp was cut, exposing the skull and the region to be drilled. The skull was cleaned and dried with sterile cotton swabs for better adhesion of the glue. The 2.5 cm x 0.5 cm, 350 mg weight aluminum headbar was placed directly on the wet glue and then dental acrylic was added to cover the glue and cement the bar in the desired position. The metal bar then stuck rigidly to the skull, without going through it. The bar was attached to the apparatus during behavior. With the aid of a mouse brain atlas, the desired position relative to the bregma in the left hemisphere (for piriform cortex AP 3.2mm; ML 3mm) was reached using a stereotaxic and a point was marked. We also implanted the ground and the reference electrodes (silver wires) in the cerebellum. After the operation, animals recovered on the

heating pad until awakening from anesthesia. We have determined that within two hours after the operation mice wake up, walk, drink, eat and do not show behavior indicative of pain sensation. However, preventively, the analgesic Tramadol was administered in a subcutaneous dose (5 mg/kg body weight), and in the drinking water. In addition, an anti-inflammatory, Ketoprofen, was administered by subcutaneous injection (5mg/kg body weight) one dose, for two days. Animals were allowed to recover from surgery for 3 days.

Neural recordings

In different training sessions, depending on the experimental group to which the animal belongs (first session or expert animals), the recording session was held, which consisted of presenting a sequence of different trials while recording the activity of neurons in the piriform cortex. To do this, one day before this session, the craniotomy was performed where the recording electrode was going to be acutely inserted the following day. For this, animals were anesthetized with isoflurane (2% induction, 0.5–1% maintenance). A drill was used to gently file down the bone in the marked area during the previous surgery. Then, with the help of a needle, a small "cap" was lifted, exposing a small area of the brain which was kept moist carefully using swabs moistened with saline solution, avoiding touching the brain. Finally, the exposed portion was covered with a thin layer of cyanoacrylic glue (WPI). At the end, 100 µl of an anti-inflammatory, Ketoprofen, was administered by subcutaneous injection (5mg/kg body weight), and then animals were allowed to wake up and recover. The next day, the awake animal was placed in the experimental device, the glue/gel covering the craniotomy was removed with forceps, and an array of recording microelectrodes (Silicon probes, Neuronexus) attached to a micromanipulator was inserted. With the aid of the micromanipulator we reached the piriform cortex, descending slowly (1 micron/second) to the target area (DV 3.8-5.2mm, depending on the animal). Once we arrived, we

waited 20 minutes to stabilize the recording position. The recording probe was previously painted with a dye (DiI D3911, Thermofisher). All recordings were performed using A1x32-Poly3-5mm-25 s-177 silicon probes (177 μm^2 site surface area, 3-column honeycomb site geometry with 18 μm lateral and 25 μm vertical site spacing, 36 μm center-to-center horizontal span, 275 μm center-to-center vertical span, 114 μm maximum shank width near the sites, 15 μm shank thickness) with an H32 connector (NeuroNexus Technologies). The animal performed the behavioral task while the neuronal activity was recorded. Once the recording session was complete, the animal was placed in its housing cage with water ad libitum. The next day, it was anesthetized intraperitoneally with Ketamine (100mg/kg body weight) and Xylazine (10mg/kg body weight). Once anesthetized, without reflexes and in an unconscious state, the animal was decapitated and the brain dissected. The animal's brain was used for verification of the location of the recording probe (Fig. 3a). To do this, the brain was cut into 50-micron slices on the cryostat, the slices were stained in DAPI preparations to visualize cell nuclei under a fluorescence microscope, and the location of the probe tip was revealed by the fluorescence of the dye DiI.

Data acquisition

Electrophysiological signals were acquired with a 32-site polytrode acute probe (A1 $\times$ 32-Poly3-5mm-25s-177, Neuronexus, MI) connected to an Acute Smartlink32 headstage (Neuronexus, MI). Unfiltered signals were digitized at 30 kHz at the headstage and recorded by a Smartbox multichannel data acquisition system (Neuronexus, MI). Experimental events and respiration signals were acquired at 30 kHz by analog and digital inputs of the Smartbox system. Respiration was monitored with a microbridge mass airflow sensor (Honeywell AWM3100V, NJ) positioned directly opposite the animal's nose. Negative airflow corresponds to inhalation and negative changes in the voltage of the sensor output.

Behavior was automatically monitored by a behavioral measurement system Bpod (Sanworks,

NY), which was programmed to control task flow according to the protocol (see *Behavioral Task*) and sent behavioral timestamps using BNC signals to the Smartbox system for synchronization with neuronal signals.

Animal displacement on the running wheel was measured with an optical rotary encoder (H5-360-IE-S, US Digital, WA) attached to the wheel axle. When the animal moved the encoder sent TTL pulses to the Smartbox (for posterior synchronization with neuronal signals), and to a microcontroller (MEGA 2560, Arduino, NY) programmed to calculate animal position. The microcontroller transmitted the positional information to the Virtual Reality MATLAB Engine (ViRMEn). It also sent TTL pulses to the Bpod to signal context zone entry (to trigger odor stimulation) and exit (to signal entry to reward zone).

Data Analysis and Statistics

All data analysis and statistical tests were performed with custom-written software using MATLAB (Mathworks). Statistical significance was assessed with the Wilcoxon rank sum test, unless otherwise noted.

Analysis of task performance

Task performance of the animals was quantified using Signal Detection Theory, SDT^{58,59}. Briefly, through analysis of the Hit rate and False-Alarm rate of a subject performing a discrimination task, SDT allows to disambiguate the contributions of the perceptual strength of the stimulus and the subject's bias to respond.

The parameter called d' (also d -prime or sensitivity) measures the perceptual strength of the target stimulus, that is the discriminability between the subject's perceptual representations associated to the presence or absence of the target stimulus:

$$d' = Z(\text{Hit rate}) - Z(\text{FA rate})$$

where Z is the normal inverse cumulative distribution function, and Hit_{rate} and FA_{rate} are the Hit

and False-alarm rates, respectively. In our task, the target stimuli are the O_{RCR} trials. A value $d' = 0$ is chance performance, and higher values of d' indicate better performances. For example, for $Hit_{rate} = 95\%$ and $FA_{rate} = 5\%$ one obtains $d' = 3.29$ (dotted line in Fig. 1C).

The parameter called *criterion* (also c) is related to the subject's bias response:

$$criterion = -0.5 \times [Z(Hit\ rate) + Z(FA\ rate)]$$

According to SDT, the subject decides to respond or not in a given trial by imposing a threshold: if the perceptual representation in the trial is larger than the threshold *criterion* then the subject responds with a "GO". Negative criterion values indicate a bias towards GO responses, positive values indicate a bias towards NO-GO responses, and *criterion* = 0 indicates no bias (neutral response).

In our task, initial animal training sessions had negative *criterion* and low d' values, indicating almost chance performance and a bias towards licking across all trials (since animals were thirsty and there was no punishment for False Alarms). As training progressed, animals steadily increased d' (increasing performance) and brought *criterion* towards a neutral unbiased response (*criterion* $\rightarrow 0$).

Processing of respiration recordings

We took an approach similar to previous studies⁶⁰. Respiration traces sampled at 30 kHz along with the neural recording signals were smoothed with a second-order Savitzky-Golay filter in 100 ms frames and the result was locally detrended by subtracting its 1-second-long median-filtered signal. The start of inhalation was defined as zero-crossings before large negative peaks in the smoothed, detrended signal. Odor inhalations were defined as inhalations happening during a 1-second long time window starting immediately after the arrival of the odorant pulse.

Spike Sorting

Extracellular voltage traces were preprocessed with common median referencing (subtraction of

the median across all channels at each time sample to remove artifacts), spike sorted using Kilosort⁶¹ (<https://github.com/cortex-lab/Kilosort>) and the obtained result was manually curated with the phy GUI (<https://github.com/kwikteam/phy>). During manual curation all clusters of putative spike events detected by each template were inspected and evaluated according to a number of criteria. First, we discarded clusters that were considered noise if their events had near-zero amplitude or if event waveforms were non-physiological and/or extended across all recording channels. Clusters containing inconsistent waveform shapes or large number of refractory period violations (<2 ms) in the autocorrelogram were also discarded. In a final step we merged pairs of clusters that had similar waveforms, showed refractory period cross-synchronization or temporally coordinated cross-correlograms that indicated a bursting neuron. Units that passed these criteria were labeled as single units and considered in our study.

Single-neuron responses

For plotting and analysis of neuronal responses we used peri-stimulus time histograms (PSTHs) that were temporally smoothed with a gaussian filter (s.d., 30 ms). These “event PSTHs” described the neuronal spiking rate temporal evolution around the timing of either the onset of sensory stimuli, task events or behavioral events (examples in Fig. 3 and Extended data Fig. 3). For describing average responses along a complete trial we constructed PSTHs that were obtained by stitching together the “event PSTHs” corresponding to the following sequence of trial events: visual context entry, first odor inhalation, animal’s response, and trial outcome (labeled as “C”, “O”, “L” and “R” in Figs 1e-f, 2c and 3d). The “animal’s response” trial event refers either to the timing of the first animal lick after odor delivery (in GO trials) or to the timing at which the animal leaves the visual context (in NO-GO trials). The “trial outcome” trial event refers to the timing of reward delivery (in HIT trials) or to the timing that corresponded to an additional median inter-lick time interval after the “animal’s response” trial event (in MISS,

FALSE ALARM and CORRECT REJECTION trials; the rationale for this is that rewards were delivered in the second lick performed in the reward zone). Each one of these 4 “event PSTHs” (C, O, L and R) were aligned to each event median time across trials, relative to first odor inhalation (for first session data C, O, L and R were aligned to -0.46s, 0s, 2.45s and 2.99s, respectively; for expert session data C, O, L and R were aligned to -0.43s, 0s, 1.38s and 1.96s, respectively). The final PSTH that described average activity of a neuron along a full trial was obtained by stitching together this sequence of time-aligned “event PSTHs” by averaging periods where there was temporal overlap between them. We confirmed that this stitching procedure did not distort the shape of the individual “event PSTHs” and precisely described the sequence of firing rate variations along the trial. All stitched PSTHs are shown in Fig. 1e, f and Extended data Fig. 8 and (where the mean firing rate of each neuron was subtracted to display variations in firing rate around the mean).

Principal Component Analysis (PCA) and Demixed PCA (dPCA)

Dimensionality reduction of the population activity data was performed with PCA or dPCA²⁹. The dPCA analysis was performed with the Matlab implementation provided by the authors (<http://github.com/machenslab/dPCA>). Details on dPCA can be found in the original publication, here we give a brief outline and describe how we applied dPCA to our data.

The data was organized by pooling all the full-trial stitched instantaneous firing rates across recordings and stimulus conditions, resulting in one array X for each training stage (i.e., first-session or expert-session stage). The dimensions of X were $N \times C \times O \times T \times K$, where N is the total number of neurons across recordings, $C=2$ is the number of contexts, $O=2$ is the number of odors, T is the number of firing rate time samples, and K is then number of trials. Thus, for each training stage, the array X has N rows in which the i -th row contains the instantaneous firing rate of the i -th neuron for all stimulus conditions and all trials (the firing rates were centered, i.e., with row

means subtracted). By averaging X over all K trials for each neuron, odor, and context, we obtained
 a matrix $\bar{X}$ of dimensions $N \times C \times O \times T$ collecting the centered full-trial stitched neuronal PSTHs.
 For PCA, the 4-dimensional array $\bar{X}$ was reshaped into a matrix $\bar{X}_{reshape}$ with N rows and M
 columns, where $M = C \cdot O \cdot T$ (that is, concatenating all stitched PSTHs conditioned on odor-
 context combinations), and singular value decomposition was applied on $\bar{X}_{reshape}$ to obtain the
 matrix W of principal components. Trajectories through principal component space were obtained
 performing the projection $\bar{X}_{reshape}^T * W$, where T refers to the transpose, reshaping this projection
 back to the 4-dimensional $N \times C \times O \times T$ form and plotting only the first 3 rows that correspond to
 trajectories through the top 3 principal components (Fig. 2b).
 The aim of dPCA is to reduce the dimensionality of the data, while obtaining latent components
 that “demix” task parameters. For this, the single-trial data array X was reshaped into a $X_{reshape}$
 with N rows and P columns, where $P = C \cdot O \cdot T \cdot K$. When shaped this way, $X_{reshape}$ can be linearly
 decomposed into a set of 4 marginalizations x_ϕ : one condition-independent term, and 3 additional
 terms containing firing rate variations specifically related to either odorant identity, visual context
 identity, or the interaction between odorant and visual context identities. Each of these
 marginalizations x_ϕ can be understood as class means of the data, associated with the task
 parameter Φ . Importantly, these terms are all uncorrelated, thus the covariance matrix
 $X_{reshape} X_{reshape}^T$ can also be linearly decomposed into the sum of covariance matrices $x_\phi x_\phi^T$
 corresponding to each of the 4 individual marginalizations. dPCA thus seeks to find separate
 encoding and decoding matrices that, when applied to $X_{reshape}$, they minimize the least-squares
 reconstruction error of each individual x_ϕ simultaneously. It can be shown that this minimization
 is a reduced-rank regression problem that can be analytically solved by SVD (hence its relation to
 PCA). This dimensionality reduction step allows to infer a few latent components, also referred as
 decoding axes, that “demix” each task parameter and can be used as linear classifiers to properly

decode the corresponding parameter throughout time in single trials (Fig. 2c and Extended data Fig. 2). Cross-validation was used to measure the time-dependent trial classification accuracy, and the significance of this accuracy was assessed by comparing performance against a shuffled set of trials (black horizontal bars in Fig. 2c and Extended data Fig. 2c, d indicate periods of significant classification accuracy). To avoid overfitting with dPCA we included the regularization term controlled by a parameter chosen through cross-validation on each dataset (results obtained without regularization were qualitatively similar).

In order to decompose data over a specific set of parameters, dPCA requires the data to contain trials for all possible parameter combinations and all neurons. Therefore, we could not further decompose our data over a decision (GO / NO-GO) parameter, since some parameter combinations were not available in the recordings (e.g., across expert-session recordings only one animal produced NO-GO responses in O_{RCR} trials and that same animal did not produce GO responses in O_{UCU} trials). Nevertheless, clear decision-related information could still be observed in our dPCA data decomposition, particularly in expert recordings, as departures of O_{RCR} trajectories from the rest of the conditions around the animal response time. This was possible since 90.3% of GO responses (243 out of 269) were concentrated in O_{RCR} trials when animals become experts.

Estimation of signal variance and dimensionality of odor responses

To estimate the fraction of total signal variance in the recorded PSTHs (dashed lines in Extended data Figs 2a,c,f), we followed the approach described by Kobak and colleagues in the dPCA formulation (Kobak 2016). The rationale is that, due to the the finite amount of recorded trials, the estimate $\bar{X}$ must differ from the “true” underlying PSTHs, and the total variance in $\bar{X}$ must contain contributions from residual noise in the trial-averaged firing estimates.

The total variance in $\bar{X}$ can be calculated as:

$$\|\bar{X}\|^2 = \sum_{n,o,c,t}^{N,O,C,T} (\overline{X_{n,o,c,t}} - \langle \overline{X_{n,o,c,t}} \rangle_{o,c,t})^2$$

where n, o, c, t are subindexes for neuron number, odour type, context type and time sample, respectively, and angle brackets indicate averaging over the corresponding variables.

The estimate of the average noise variance across conditions for neuron n is related to the sum of the square differences between firing rates in each trial and the trial averaged firing rates:

$$\widetilde{C}_n = \sum_{k,o,c,t}^{K,O,C,T} \frac{(X_{n,o,c,t,k} - \overline{X_{n,o,c,t}})^2}{trialNum_{n,o,c}}$$

where k is a subindex for trial number, and $trialNum$ indicates the total number of trials in each odour and context condition.

Thus, the total residual noise sum of squares is:

$$\Theta = OCT \sum_n^N \frac{\widetilde{C}_n}{\overline{trialNum}_n}$$

Where

$$\overline{trialNum}_n = \frac{1}{OC} \sum_{o,c}^{O,C} trialNum_{n,o,c}$$

is a rebalancing term for pooling together sequentially recorded neurons with different total number of trials between conditions.

Finally, the total signal variance is

$$SigVar = \|\bar{X}\|^2 - \Theta$$

and the dashed lines in Extended data Fig. 2a,c,g report the fraction of total signal variance, which is $100 (1 - \Theta / \|\bar{X}\|^2)$.

To characterize the dimensionality of odor responses we estimated the total number of principal

components needed to explain the fraction of total signal variance (Extended data Fig. 2g). The fraction of total signal variance in the population of responses aligned to odour stimulation was higher for expert animals, indicating that odor responses are more reliable after learning. This was true even when matching the total number of trials in both training states: 71.7% and 55.4% total signal variance for experts and first sessions, respectively (dashed lines in Extended data Fig. 2g). To confirm the increase in reliability in the population of odor responses after learning, we calculated the coefficient of variation to estimate trial-to-trial variability in PCx neuronal firing rates. Odour stimulation reduced the average coefficient of variation, as previously reported in rats^{5,28}, but the reduction was stronger for expert animals (coefficient of variation of spike counts across trials: 1.70 +/- 0.05 for first sessions and 1.8+/-0.04 for experts, mean +/- s.e, p=0.02, Extended data Fig. 2f). This supports the lower total signal variance of odor responses observed in first session animals.

Generalized Linear Model (GLM) for encoding and decoding analysis

For this analysis the spiking-activity time series were discretized into 10-ms bins. The analysis was performed by adapting the Matlab implementation of Poisson GLM regression for trial-based spiking data by Park and colleagues³³ (<https://github.com/pillowlab/neuroGLM>).

This framework poses an encoding model $p(\mathbf{r}|\mathbf{x})$ of the probabilistic relationship between a neuronal response $\mathbf{r}$ and a set of task variables $\mathbf{x}$ in a given trial. The probability of response is related to an underlying time-varying spiking rate λ_t , which is given by:

$$\lambda_t = \exp \left(\sum_i (k_i \otimes x_i)_t \right)$$

399

400 where $x_i(t)$ is the time course of i th variable in the model, k_i is the neuron’s “kernel” (i.e., linear
 401 filter) associated with that variable, and $\otimes$ indicates a linear convolution operation. The kernel
 402 captures the relationship between this variable and the neuron’s probability of spiking. In turn, in
 403 a given single trial, the probability of spiking of the neuron is given by a Poisson distribution (i.e.,
 404 it assumes Poissonian spiking statistics):

$$405 \quad p(\mathbf{r}|\mathbf{x}, \{k_i\}) = \prod_{t=0}^T p(r_t|\mathbf{x}, \{k_i\}) = \frac{1}{r_t!} \prod_{t=0}^T (\Delta\lambda_t)^{r_t} e^{-\Delta\lambda_t}$$

406 where Δ is the time bin size (10 ms), T is the total number of time bins in the trial, and r_t is the
 407 spike count at time bin $(t, t+\Delta)$.

408 *Parametrization of model variables and kernels:*

409 A schematic representation of the parametrization is shown in Extended data Fig. 5a. This choice
 410 of regression variables was based on the neuronal response modulations observed in the PSTHs
 411 and in population dynamics (dPCA). In our model there were two kinds of kernels: “event-based”
 412 kernels (i.e., kernels reflecting the time-varying influence of a task variable on the time-varying
 413 spike rate), and “tuning-based” kernels (i.e., kernels describing the dependence of neuronal firing
 414 on the value of a task variable). Kernels were parametrized by a set of basic functions (Extended
 415 data Fig. 5b). For any kernel parametrized by n bases, the model infers a set of n coefficients that
 416 correspond to the weighted linear combination of the n basis functions. To choose the number of
 417 bases n used to parametrize each kernel, we ran a series of preliminary tests on a set of neurons
 418 and evaluated parametrization performance through cross-validation (see *Model fitting and model*
 419 *performance*).

420 *i. Event-based kernels*

421 Each event associated with the variable was represented as a delta function over time, and

convolved with the corresponding kernel (Extended data Fig. 5a). To parametrize the associated event kernels, we used basis functions defined by a series of raised cosine bumps centered at different times spanning the range of time covered by the kernel (Extended data Fig. 5b).

- Inhalation onset: 10 bases covering 460 ms window after the onset of each inhalation. Labeled “Inhal”.
- Inhalations of rewarded and unrewarded odors: 10 bases covering 460 ms window after the onset of each odor inhalation during odor stimulation. One kernel for each odor, labeled “O_R” and “O_U”.
- Licking: 15 bases covering 360 ms window after the onset of each lick. Labeled “Licks”.
- Reward consumption: 15 bases covering 360 ms window after the onset of each lick. Labeled “Reward”.
- Anticipation to GO response: 22 bases covering 2000 ms before the first lick that follows odorant stimulation. Labeled “preGO”.
- Modulation of odor responses by presence of rewarded context: 20 bases covering 1500ms after the first odor inhalation. One kernel for each odor, labeled “modO_R” and “modO_U”. These kernels aim at describing the interaction between odors and contexts.

ii. Tuning-based kernels

Each variable was discretized and represented as an animal-state vector for the variable at a given instant in time, and convolved with the corresponding kernel (Fig. 5a). Each animal-state vector denotes a binned variable value, all of whose elements are set to 0, except for one element, which is set to 1, corresponding to the bin the current animal state occupies. To parametrize these tuning

kernels, we used basis functions defined by a series of raised cosine bumps centered at different variable values along the range spanned by the animal-state vector (Extended data Fig. 5b).

- Animal position along rewarded and unrewarded visual contexts: 42 bases covering the entire length of the virtual corridor discretized in a sequence of 4-cm bins. One kernel for each context, labeled “C_R” and “C_U”.
- Running speed: a sequence of bins of 1 cm/s up to the animal’s maximum running speed (the total number of bases varied between animals depending on their maximum speed, typically ~35 bases were used). Labeled “Speed”.

iii. Bias kernel

To account for the average firing rate of the neuron during the experiment, we introduced a bias kernel parametrized by a single coefficient that was convolved with a boxcar function that lasted for the whole recording (that is, the bias kernel was constantly active). This kernel is not shown on Extended data Fig. 5a.

Model fitting and model performance:

To fit the weights of the basic functions that parametrize the kernels in the GLM encoding model of each neuron, we maximized the model’s log-posterior, using a ridge prior to regularize the inferred weights (under the assumption that kernels should be relatively smooth):

$$L(\mathbf{k}) = \sum_{t=0}^T (r_t \log(\lambda_t \Delta) - \lambda_t \Delta) + C - \xi ||\mathbf{k}||^2$$

where $\mathbf{k}$ is a vector representing the weights to be obtained, C is a constant that does not depend on $\mathbf{k}$ (thus irrelevant for the optimization procedure), and ξ is the smoothing hyperparameter controlling regularization.

Model performance was quantified through cross-validation, by computing the model log-likelihood (the log-posterior without the last regularizing prior term) of held out data under the model. For this, we randomly divided trials into 10 sets of equal size. The cross-validation procedure was repeated on each of these 10 sets of trials (10-fold cross-validation; the log-likelihood of the model on each fold was normalized by the number of spikes in the fold). This procedure penalized model overfitting, thus allowing valid performance comparisons between models of varying complexity.

To select the hyperparameter ξ we took an Empirical Bayes approach, in which we maximized the marginal likelihood (also called model evidence) with respect to ξ (chosen from a grid) and used this estimate of ξ into the posterior. To compute the marginal likelihood we used the Laplace approximation. This procedure does not require cross-validation, and it was performed on all trials at once.

Once a model was selected (see *Model selection*), the final estimate of $\mathbf{k}$ was inferred by fitting the model on all trials at once.

Model selection:

Three variables were associated with 2 kernels that depended on the identities of odorants and/or contexts: inhalations of rewarded and unrewarded odors (O_R and O_U), modulation of odor responses by presence of rewarded context ($modO_R$ and $modO_U$), and position along rewarded and unrewarded visual contexts (C_R and C_U). When regressing neuronal responses against these 3 variables, the 2 corresponding kernels were used. Thus, the total number of regression variables was 8, and the models considered included combinations between all of them.

Testing all combinations of variables is intractable. Thus, we implemented a forward-search model selection procedure similar to the one introduced by Hardcastle and collaborators⁶², that selects

the simplest model that best predicts held out neuronal activity data (spike-normalized model log-likelihood on 10 cross-validation data folds; see *Model fitting and model performance*). First, models including a single variable were evaluated, and the one with highest performance was determined. This single-variable model was then compared against all models with two variables that included this single variable. If the double-variable model with highest performance was significantly better than the best single-variable model, we proceeded to compare this double-variable model to the best performing triple-variable model that included this pair of variables. We continued including variables in the model through this procedure until the more complex model considered did not show an improvement in prediction performance, in which case the simpler model was selected. Significance in performance improvement was always assessed with a two-sample one-sided signed rank test, which tests the hypothesis that the difference in performance across data folds between the more complex and the simpler model comes from a distribution with median greater than 0. Models were considered better performing if they had a p -value lower than 0.05 under this test. Neurons for which the selected model did not perform significantly better than a model with constant mean firing rate were considered as not modulated by task variables (i.e., they had 0 modulating variables).

Validating the exponential nonlinearity of GLM models:

The GLM model assumes an exponential nonlinearity that governs the mapping from the kernel output (the result of the linear convolution $k \otimes x$) to the time-varying spiking rate λt . To assess if this is a valid assumption, we ran the GLM fitting procedure for a given neuron, binned the kernel output across the recorded trials and calculated the observed spike rate for each of these bins, along with the result of applying an exponential non-linearity over the kernel output bins. This analysis showed that the exponential non-linearity is an adequate approximation across the range of kernel

output bins observed in the data, with some deviations in extreme and under-sampled bins (see neuron examples in Extended data Fig. 7a).

Evaluating the kernel estimation accuracy of GLM models:

GLM encoding models aim to quantify the dependencies of the neural response on multiple coexistent (stimulus, behavioral and task-related) variables. As pointed out before³³, randomness and variability in the timings of these experimental variables are essential for disentangling the different components of the neuronal response. If the relative timings of a given pair of variables was constant across trials, it would be impossible to dissociate the temporal evolution of the firing rate components associated with each of these two variables. For example, in a GO-NOGO task where the animal can collect a reward at a fixed short time after being stimulated with an odor, one cannot disentangle the temporal course of a pre-GO neuronal modulation from the odor response component. In the present virtual reality task, the timing of events is under operant control of the animal, resulting in important timing variability throughout trials. Nevertheless, some degree of temporal correlation between variables could remain, impacting negatively in the ability of the model to discriminate between the contributions of these variables.

To assess the effectiveness of our GLM model in quantifying the contributions of variables to the neuronal response given the recorded data during the task, we used simulated data.

Using an artificial set of known kernels $\mathbf{k}^s$ and the set of task variables recorded during real trials $\mathbf{x}$, we simulated a spike-count dataset $\mathbf{r}_t^s$ that we regressed with the GLM encoding model, and evaluated the accuracy of the model's estimates of the kernels, $\hat{\mathbf{k}}^s$. This would allow us to determine the effectiveness of both the kernel inference procedure and the statistical framework for model selection, when the model is faced with the recorded task data $\mathbf{x}$ (and taking into account the task design).

To simulate the spiking rate of a neuron λ_t^s , first we convolved the recorded $\mathbf{x}$ (also called “design matrix”) with a random combination of kernels $\mathbf{k}^s$ (by randomly choosing a set of kernels from the pool of kernels inferred when analyzing real recorded spike-count data), and applied an exponential non linearity to the output of the convolution:

$$\lambda_t^s = \exp\left(\sum_i (k_i^s \otimes x_i)_t\right)$$

We then simulated $\mathbf{r}_t^s$ as a time series of 10-ms-binned spike-count dataset by sampling from a Poisson distribution with parameter λ_t^s . We simulated 100 neuronal responses from task data $\mathbf{x}$ of both first-session and expert-session recordings.

Using $\mathbf{r}_t^s$ we performed the cross-validated model selection procedure described before to obtain the $\hat{\mathbf{k}}^s$ estimates, which we compared with $\mathbf{k}^s$. The correlation between $\mathbf{k}^s$ and $\hat{\mathbf{k}}^s$ was high across all neuronal simulations (Extended data Fig. 7b; median correlation of 0.95 and 0.951 for simulations based on first-session and expert-session data), and the correlation between the multiplicative neuronal gain of individual $\mathbf{k}^s$ and $\hat{\mathbf{k}}^s$ coefficients was also high (Extended data Fig. 7c). Inspection of individual kernels showed that the simulated kernels were accurately estimated (Extended data Fig. 7d), and discrepancies in which kernels were not detected by the kernel estimation procedure (i.e., included in $\mathbf{k}^s$ simulations but missing in $\hat{\mathbf{k}}^s$) or erroneously estimated (i.e., not included in $\mathbf{k}^s$ simulation but estimated in $\hat{\mathbf{k}}^s$) were typically for kernels of multiplicative gain ~ 1 which have negligible impact on neuronal response encoding (see kernels labeled with asterisks in (Extended data Fig. 7d). Altogether these results indicate that the model can accurately estimate kernels from the recorded data, capturing with precision the individual contributions of stimulus, behavioral and task-related variables to neuronal responses.

Contribution of a task variable to the GLM encoding model:

We estimated the contribution of a given variable x_i (associated with a kernel k_i) to the encoding model of a neuron by calculating the difference in model performance (spike-normalized log-likelihood increase) between the selected model θ (with firing rate λ_t) and the model $\theta^{\sim k_i}$ (with firing rate $\lambda_t^{\sim k_i}$) that contains all variables in the selected model, minus the variable x_i :

$$\text{Contribution of variable } x_i = \frac{\sum_{t \in \{t_{k_i}\}} [(r_t \log(\lambda_t \Delta) - \lambda_t \Delta) - (r_t \log(\lambda_t^{\sim k_i} \Delta) - \lambda_t^{\sim k_i} \Delta)]}{\log(2)(1 + \sum_{t \in \{t_{k_i}\}} r_t)}$$

Here the sum is performed over the set of time bins $\{t_{k_i}\}$ in which the convolution ($k_i \otimes x_i$) was performed, that is the moments in time where the kernel k_i contributed to the firing rate λ_t . The normalizing factor has an offset of 1 in the sum of r_t over $\{t_{k_i}\}$ to avoid numerical undefined expressions when no spikes were emitted during $\{t_{k_i}\}$, and the $\log(2)$ is included to measure the spike-normalized log-likelihood increase in terms of bits. The *Contribution of variables* calculated for all recorded neurons is shown in Extended data Fig. 5d. For the percentual *Relative contribution* of each variable (Fig. 4c), for each neuron we normalized each variable contribution by the sum of all contributions:

$$\text{Relative contribution of variable } x_i = \frac{\text{Contribution of variable } x_i}{\sum_i \text{Contribution of variable } x_i}$$

Decoding with GLM model:

Decoding with the fitted GLM models, allows trial decoding of binary variable categories, while taking into account neuronal response modulations related to all the other task variables. As was previously shown⁷, under this GLM model, the computation for decoding the identity of a variable x_j with two categories $x_j = A$ and $x_j = B$ (e.g., rewarded or unrewarded odor) with kernels k_j^A and k_j^{AB} , respectively, is rather straightforward. Consider the log likelihood ratio (*LLR*) for this situation:

$$LLR = \log \frac{likelihood_{x_j=A}}{likelihood_{x_j=B}} = \log \frac{p(\mathbf{r}|\mathbf{x}^{\sim j}, x_j = A)}{p(\mathbf{r}|\mathbf{x}^{\sim j}, x_j = B)}$$

where $\mathbf{x}^{\sim j}$ is a vector with the rest of the variables other than x_j (i.e., $\mathbf{x} = \{\mathbf{x}^{\sim j}, x_j\}$). Further developing this formulation leads to:

$$LLR = \sum_{t=0}^T \left[r_t \left((k_i^A \otimes x_j)_t - (k_i^B \otimes x_j)_t \right) + \Delta(\lambda_t^A - \lambda_t^B) \right]$$

where λ_t^A and λ_t^B are the time-varying spiking rate for the model with $x_j = A$ and $x_j = B$ respectively. The LLR leads to a simple rule for decoding the category of the variable x_j : when $LLR > 0$, the recorded neuron indicates that $x_j = A$ that is more probable under the model; on the contrary, $x_j = B$ is more probable if $LLR < 0$. This mathematical procedure can be conceptualized as a time-varying estimation of whether the emitted spikes up to that time point in the trial are more probable under the spike rate prediction of a model that considers one of the variable categories rather than the alternative. When considering a population of neurons recorded simultaneously, the $LLRs$ of the individual neurons can be summed together. In Fig 5a, this decoding accuracy is averaged across animals. For the analysis of the relation between the accuracy and the number of neurons used for decoding (Extended data Fig 10a-c), pseudopopulations of neurons were constructed with subsets of neurons that were randomly sampled from across recording sessions, up to the total number of 117 neurons (matching the total number of neurons from first-session and expert-session recordings). For each population size, the sampling procedure was repeated 50 times and accuracy was averaged. For the analysis shown in Extended data Fig. 10f aimed at evaluating the contribution to odor decoding accuracy of non-olfactory information encoded in PCx neurons, we calculated $LLRs$ where A was set to O_R and B to O_U , and λ_t^A and λ_t^B were calculated in two different ways: one using the full models (i.e.,

including all fitted kernels) and another after removing contributions of non-odor-related kernels from the models of odor-responsive neurons (i.e., removing non-olfactory modulations from neuron models that had olfactory kernels).

Expert-session recordings typically contained more trials than first-session recordings. To control for the effect of this difference on model estimation and hence accuracy of decoding, we re-fitted GLM models after matching the number of trials used from both training conditions. All decoding results are shown after this trial matching procedure.

In Fig. 5b we compare $p(O_R | \text{spikes})$ and $p(C_R | \text{spikes})$ with the behavioural response delay (in GO trials the latter corresponds to the reaction time to first lick after first odour inhalation, while in NO GO trials it corresponds to the delay to leave the context zone after first odour inhalation). The quantity $p(O_R | \text{spikes})$, the posterior probability of O_R given the observed spikes, is a GLM-based decoding of the PCx representation of O_R in a trial. Following Park and collaborators (Park et al 2014), we calculated $p(O_R | \text{spikes})$ from the GLM encoding distribution using Bayes' rule. If we assume the prior probabilities of O_R and O_U are equal (which is reasonable, since both odours were presented with equal probabilities across trials), then $p(O_R | \text{spikes})$ is obtained by:

$$p(O_R | r, \{x^{\sim \text{odours}}_i\}) = \frac{p(r | O_R, \{x^{\sim \text{odours}}_i\})}{p(r | O_R, \{x^{\sim \text{odours}}_i\}) + p(r | O_U, \{x^{\sim \text{odours}}_i\})}$$

where on the right side we observe the GLM encoding models and $\{x^{\sim \text{odours}}_i\}$ is the set of all covariates except the odour ones. For $p(C_R | \text{spikes})$ a similar formula holds, replacing O_R and $\sim \text{odours}$ by C_R and $\sim \text{context}$.

Decoding odors with linear classifiers

For the odor decoding analysis of Passive vs. Task conditions (Fig. 6b) and the evaluation of the contribution to odor decoding accuracy of non-olfactory information encoded in PCx neurons (Fig. 6c), we used linear classifiers based on L2-regularized logistic regression with 3-fold cross-validation. This analysis is independent of the GLM models or GLM-based decoding. This guarantees that the obtained decoding accuracies will not be influenced by differences across conditions in the goodness-of-fit or expressive power of GLM models.

Neuronal responses were concatenated into a Trials (T) x Neuron (N) matrix, arranged so that each row contained trial responses to a particular odor. Each neuron's response was averaged over a 0.5-s window after first odor inhalation (for the Passive condition, peak decoding was reached later, at 1s, so we used that time window). Trials were labeled according to the odor sampled, and arranged into a vector of T elements.

Pseudopopulations of neurons were constructed with subsets of neurons that were randomly sampled from across recording sessions, up to the total number of 117 neurons (matching the total number of neurons from first-session and expert-session recordings). Because different trial conditions contained different numbers of trials, a random subset of responses from each neuron was excluded so that the number of trials per condition was consistent across neurons in the pseudopopulation. For each population size, this trial sampling procedure was repeated 50 times and the obtained accuracies were averaged.

For the analysis of Fig. 6c, a total number of 48 trials were used. For the Passive vs. Task conditions a total number of 25 trials were used (Fig. 6b; the fewer number of trials explains the lower performance reached in Fig. 6b compared to Fig. 6c).

Clustering groups of neurons with similar relative contributions of their encoded variables

To group neurons according to the task variables that modulate their activities, we used an

652 agglomerative hierarchical clustering approach. We only considered neurons whose GLM
 653 encoding model contained at least one task variable (i.e., with a total number of modulating
 654 variables ≥ 1 ; first session: 115 out of 177 neurons, 65.0%; expert session: 75 out of 117 neurons,
 655 64.1%). First, for each neuron k we determined the vector ***RelContrib*** _{k} =
 656 $\{\text{Relative contribution of variable } x_i\}_i$ containing relative contributions of all variables to the
 657 neuron k . We then used a correlation metric to calculate the pairwise distances between all vectors
 658 of relative contributions, $pdist_{i,j} = 1 - r_{i,j}$, where $r_{i,j}$ is the sample correlation between the pair
 659 (***RelContrib*** _{i} , ***RelContrib*** _{j}). With these distances we built an agglomerative hierarchical cluster
 660 tree, where distances between clusters were specified by the unweighted average distance between
 661 all pairs of objects in any two clusters (UPGMA). The obtained hierarchical trees are shown in
 662 Extended data Fig. 9a, d, and the vectors of relative contributions sorted according to their
 663 corresponding clusters are shown in Extended data Fig. 9b, e. To cut a hierarchical tree into
 664 individual clusters, we grouped tree leaves using a cutting threshold $thresh_{cut}$ for the distance
 665 coefficients of nodes in the tree (i.e., cutting through the tree at nodes whose heights were lower
 666 than $thresh_{cut}$; clusters with less than 3 neurons were discarded). The obtained clusters of relative
 667 contributions are shown in Fig. 4e, Extended data Fig. 9c, f with a color code that is maintained in
 668 Extended data Fig 9.

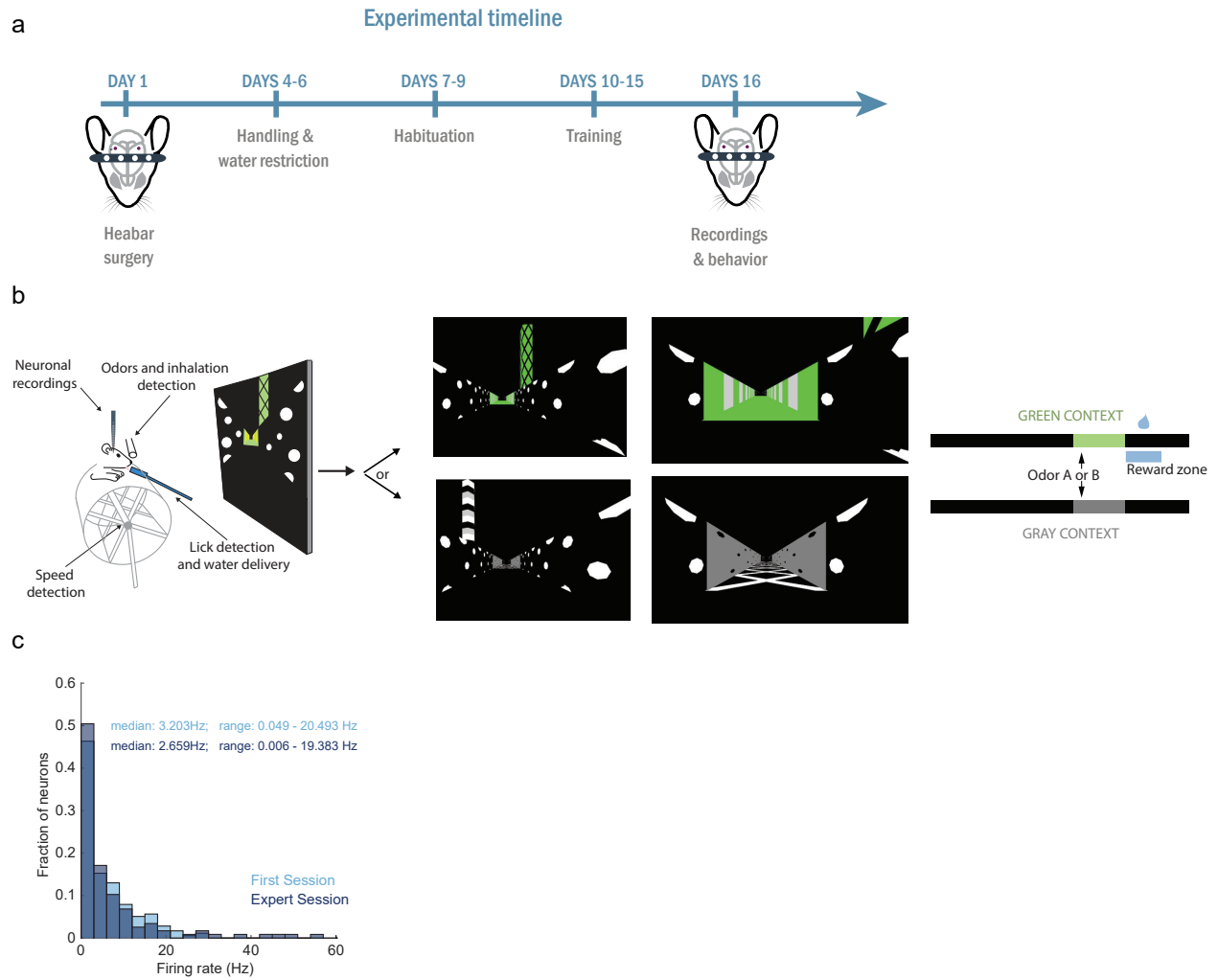
669 We determined the optimal value for $thresh_{cut}$ by cross-validation of the generalizability of the
 670 resulting clusters. In order to do this, we randomly divided trials into 2 sets of equal size (sets A
 671 and B) and separately fitted the GLM model to each trial set, obtaining the vectors of the relative
 672 contributions of all variables to neuron k under the set A and under the set B, ***RelContrib***^A _{k} and
 673 ***RelContrib***^B _{k} correspondingly. We first evaluated how well clusters in set A generalized to set B:
 674 we labeled A as the train set, and B as the test set. We then fixed $thresh_{cut}$ on a given value, and
 675 performed the described hierarchical clustering procedure for both ***RelContrib***^A _{k} and ***RelContrib***^B _{k} ,

obtaining a partitioning matrix for each set, P_{Train}^A and P_{Test}^B (if N is the total number of neurons,
 the partitioning matrix P is a binary $N \times N$ matrix, where matrix position P_{ij} is set to 1 if neuron i
 and neuron j belong to the same cluster, and set to 0 otherwise). We then constructed an additional
 partitioning matrix $P^{A \rightarrow B}$ by using the centroids of clusters in P_{Train}^A to perform clustering on set
 B : we assigned neuron k to cluster j in $P^{A \rightarrow B}$ if $\mathbf{RelContrib}_k^B$ was closest to the centroid of cluster
 j across all cluster centroids in set A . To quantify generalizability of clustering results, we
 calculated the sample correlation coefficient between the upper triangular block of matrix P_{Test}^B
 and the upper triangular block of matrix $P^{A \rightarrow B}$. We repeated this procedure using B as train set
 and A as test set, and varied $thresh_{cut}$ along a range of values. Through this procedure, we
 determined that the optimal $thresh_{cut}$ was 2.2 for first-session data, and 2 for expert-session data.
 For these values of $thresh_{cut}$, generalizability of first-session data was 0.79 +/- 0.006 (for shuffled
 test partitionings it was 0.002 +/- 0.032) and of expert-session data was 0.69 +/- 0.09 (for shuffled
 test partitionings it was 0.003 +/- 0.030).
 To visualize the organization of the relative contribution vectors of all neurons, and perform a
 qualitative validation of the obtained clusters, we used multidimensional scaling (MDS). This
 nonlinear technique maps the collection of neuron's relative contribution vectors $\{\mathbf{RelContrib}_i\}_i$
 from their original eight-dimensional space (corresponding to the 8 task variables) to a
 bidimensional projection that best preserves the original distances between vectors $\{pdist_{i,j}\}_{i,j}$.
 Neurons with similar relative contribution vectors occupy similar regions of the bidimensional
 MDS plane, allowing visualization of the global structure of similarities of $\{\mathbf{RelContrib}_i\}_i$. We
 applied the non-classical (non-metric) MDS algorithm on the collection of $pdist_{i,j}$ values,
 calculated with the euclidean distance metric, with Kruskal's normalized stress-1 goodness-of-fit
 criterion. In Fig. 4e, neurons in the MDS plane were colored according to the cluster to which they
 belong, as determined by our hierarchical clustering approach (same color-code used in Extended

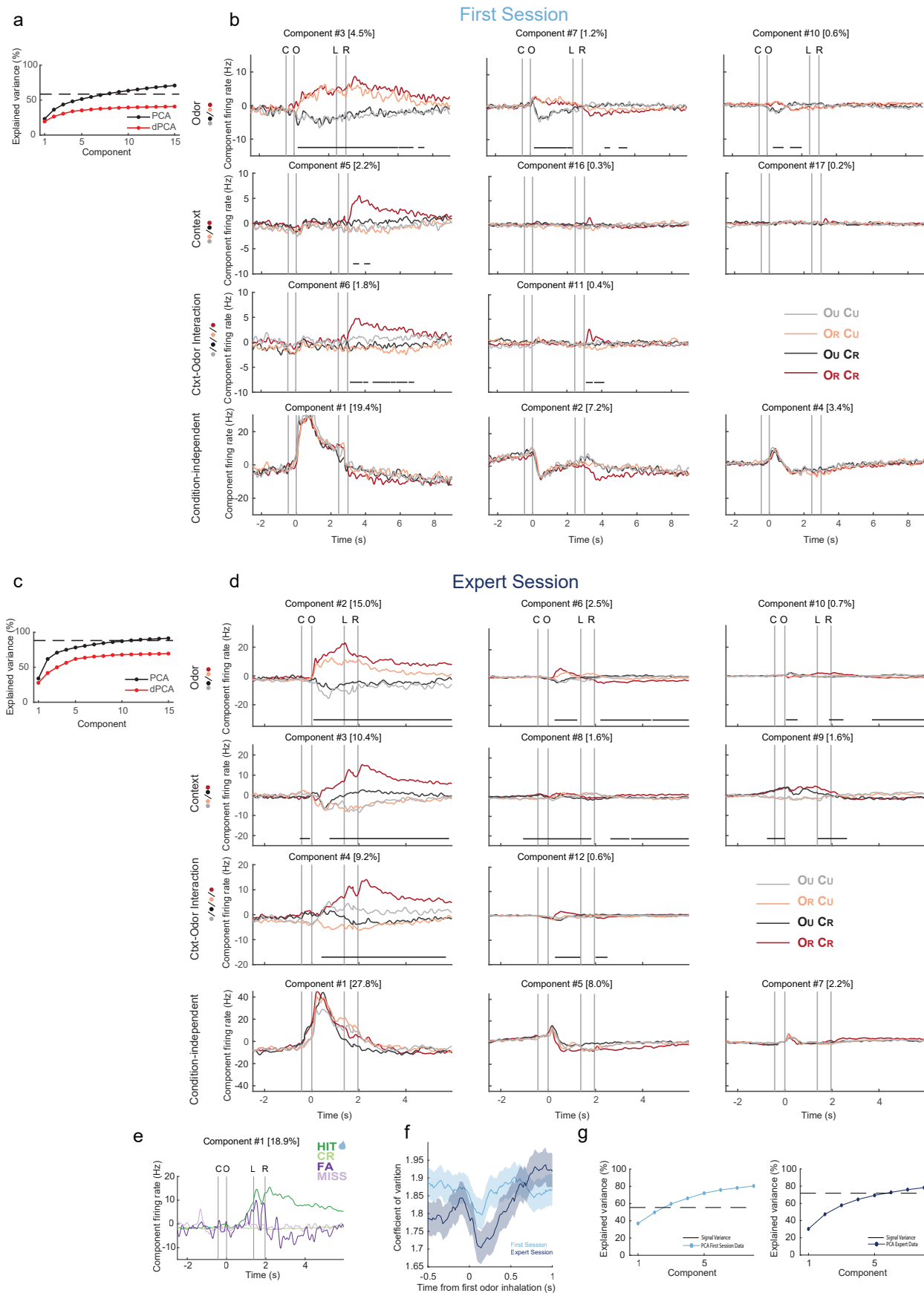
data Fig. 9).

Methods References

57. Aronov, D. & Tank, D. W. Engagement of neural circuits underlying 2D spatial navigation in a rodent virtual reality system. *Neuron* 84, 442–456 (2014).
58. Carandini, M. & Churchland, A. K. Probing perceptual decisions in rodents. *Nat Neurosci* 16, 824–831 (2013).
59. Green, D. M. & Swets, J. A. Signal detection theory and psychophysics. xi, 455 (John Wiley, 1966).
60. Bolding, K. A. & Franks, K. M. Complementary codes for odor identity and intensity in olfactory cortex. *Elife* 6, e22630 (2017).
61. Marius Pachitariu, Shashwat Sridhar, & Carsen Stringer. Solving the spike sorting problem with Kilosort. *bioRxiv* 2023.01.07.523036 (2023) doi:10.1101/2023.01.07.523036.
62. Hardcastle, K., Maheswaranathan, N., Ganguli, S. & Giocomo, L. M. A Multiplexed, Heterogeneous, and Adaptive Code for Navigation in Medial Entorhinal Cortex. *Neuron* 94, 375-387.e7 (2017).



Extended data Fig. 1. Experimental setup and timeline. **a**, Experiment timeline indicating the training and recording protocol. **b**, Experimental setup showing the two alternative virtual reality corridors (grey or green contexts) used in the experiments. **c**, Distribution of firing rates of recorded neurons for first-session recordings (light blue) and expert-session recordings (dark blue).

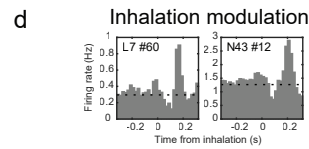
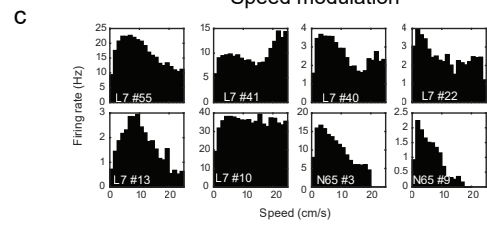
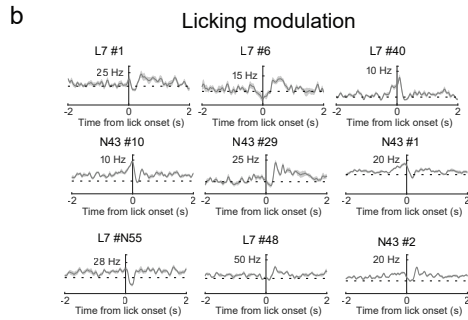
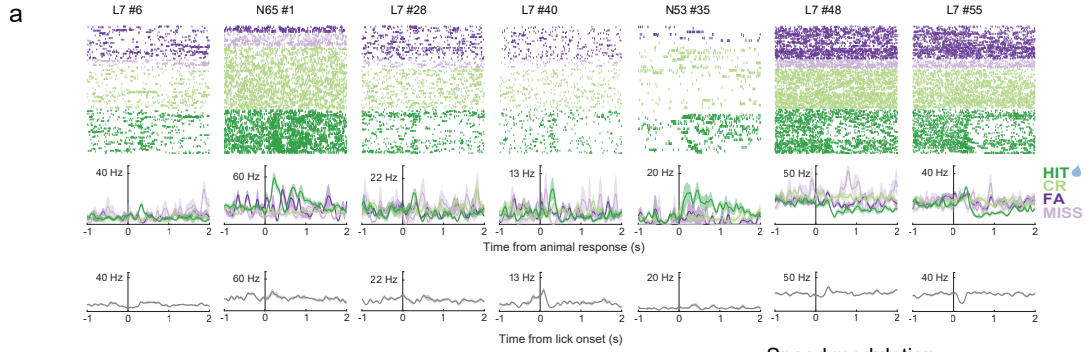


Extended data Fig. 2. Demixed PCA components. a and c, Cumulative variance explained by

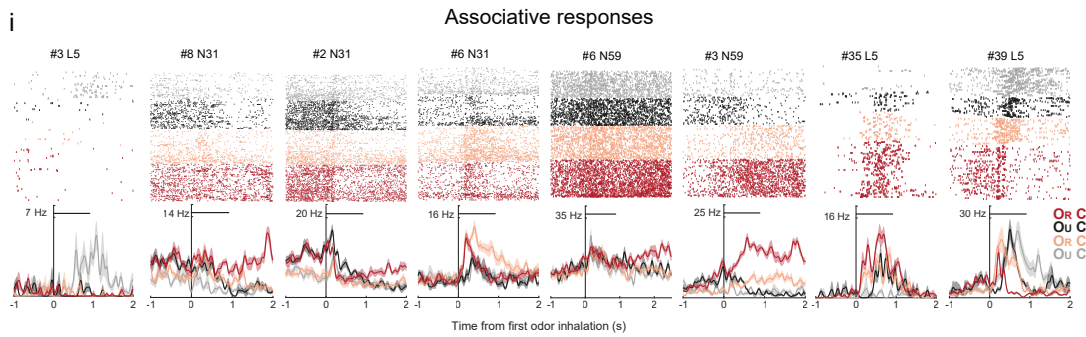
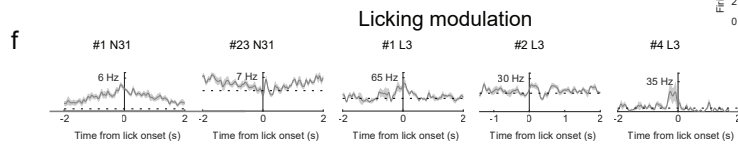
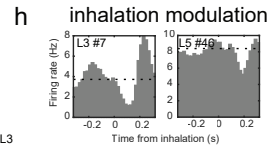
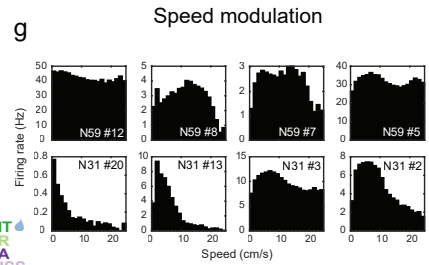
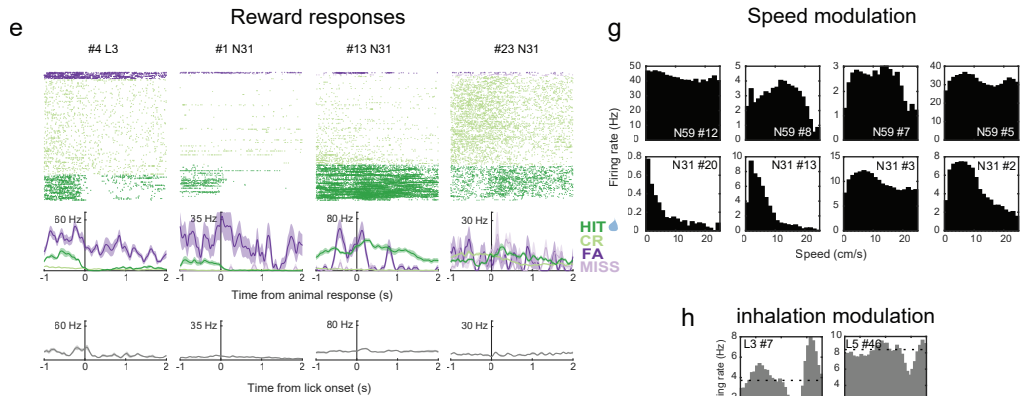
dPCA and PCA from neurons recorded of animals in first session (a) or expert session (c). Dashed line shows an estimate of the fraction of signal variance in the data. **b and d**, Demixed principal components. From top to bottom panels: odour discrimination components, context discrimination components, context-odour interaction discrimination components, condition-independent components for first session (b) of expert session (d) recordings. In each subplot, the full data are projected onto the respective dPCA decoder axis. Thick black lines show time intervals during which the respective task parameters (odour, context, and context-odour interaction) can be significantly decoded from single-trial activity (see Materials and Methods). **e**, First dPCA component of piriform activity arranged according to trial outcome in expert animals (hit and correct rejection in dark and light green, false alarm and miss in dark and light purple). Variance explained by the component is shown as percentage. **f**, Coefficient of variation for odour responses aligned to first odour inhalation onset. Trial-to-trial coefficient of variation for each neuron was calculated for different odour trials and averaged. (Coefficient of variation of spike counts across trials: 1.70 ± 0.05 for first sessions and 1.8 ± 0.04 for experts, mean $\pm$ s.e, $p=0.02$). **g**, Cumulative variance explained by PCA from odour responses of neurons recorded for animals in first session (top) or expert session (bottom). Odor responses comprise a time window of 1 second following first odour inhalation onset. Dashed line shows an estimate of the fraction of signal variance in the data (*i.e.*, the variance that cannot be attributed to noise variance due to variability across trials), that was higher for expert animals (71.7% and 55.4% total signal variance for experts and first sessions, respectively, matching total number of trials for both conditions).

First Session

Reward responses

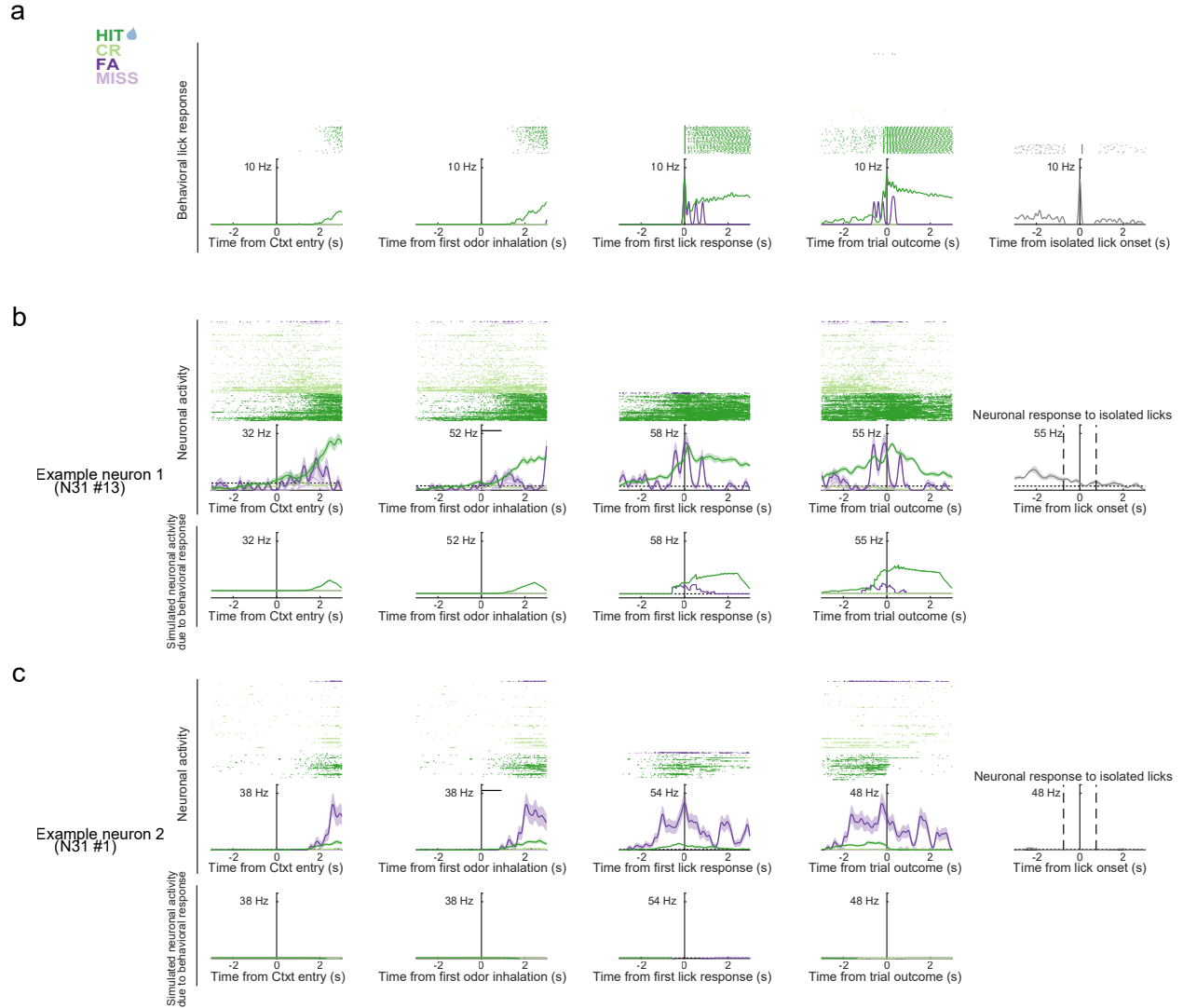


Expert Session



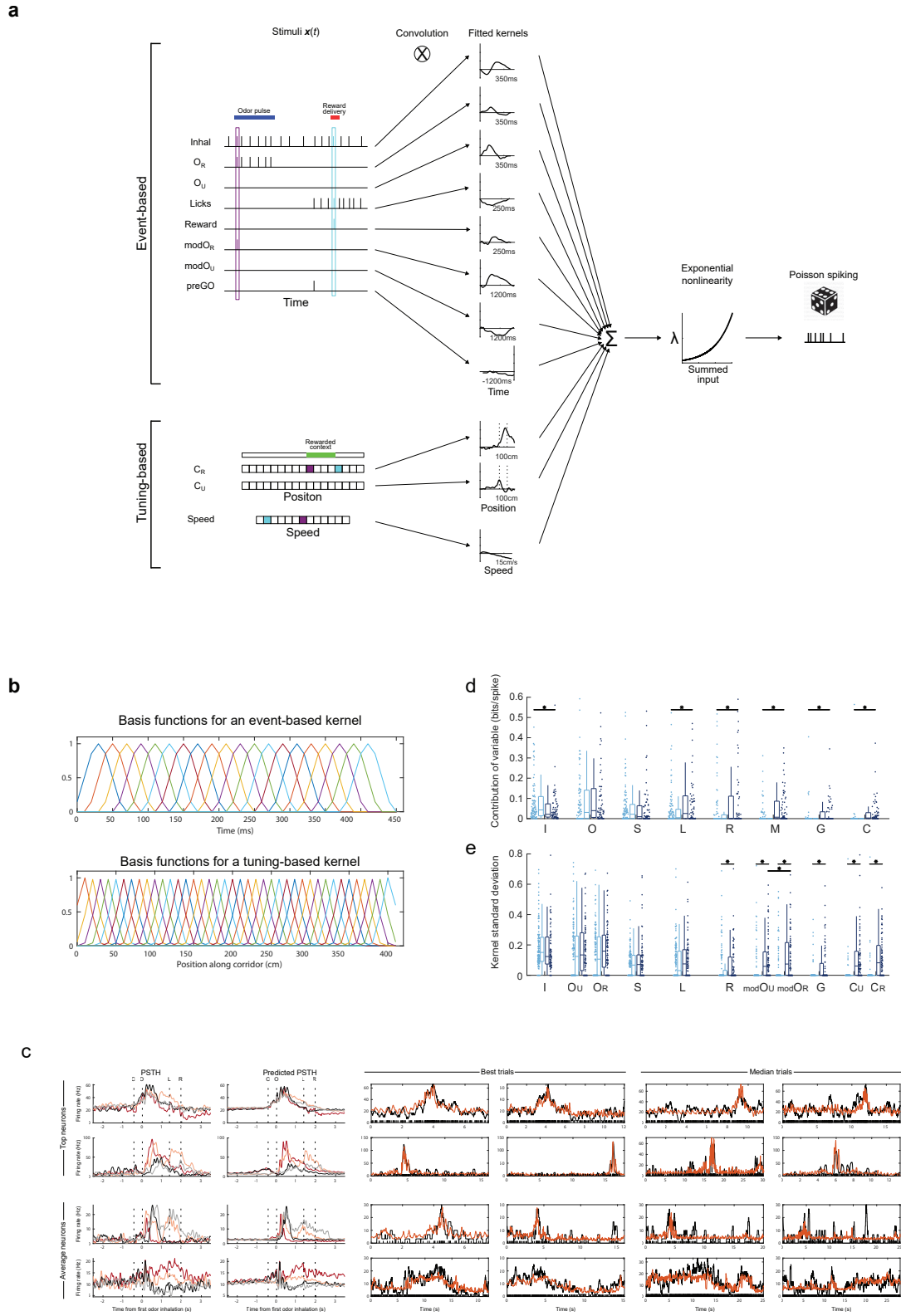
Extended data Fig. 3. a and e, Examples of reward responses are shown in raster plots of action

potentials (ticks) colour coded by trial outcome (colour labels shown on the right) for first session (a) or expert sessions (e). Middle panels show the average firing rate of each raster aligned to the animal's response after odour delivery. Bottom panels show the average firing rate of these neurons, aligned to the onset of every lick. **b and f**, Examples of responses of neurons modulated by licking in first sessions (b) or in expert sessions (f). Dashed line indicates mean firing rate. **c and g**, Examples of responses of neurons modulated by speed in first session (c) or in expert session (g). **d and h**, Examples of responses of neurons modulated by inhalation in first sessions (d) or in expert sessions (h). Dashed line indicates mean firing rate. (i) Examples of neurons with associative responses in expert animals. Horizontal black line shows odor pulse duration. Color labels shown on the right.



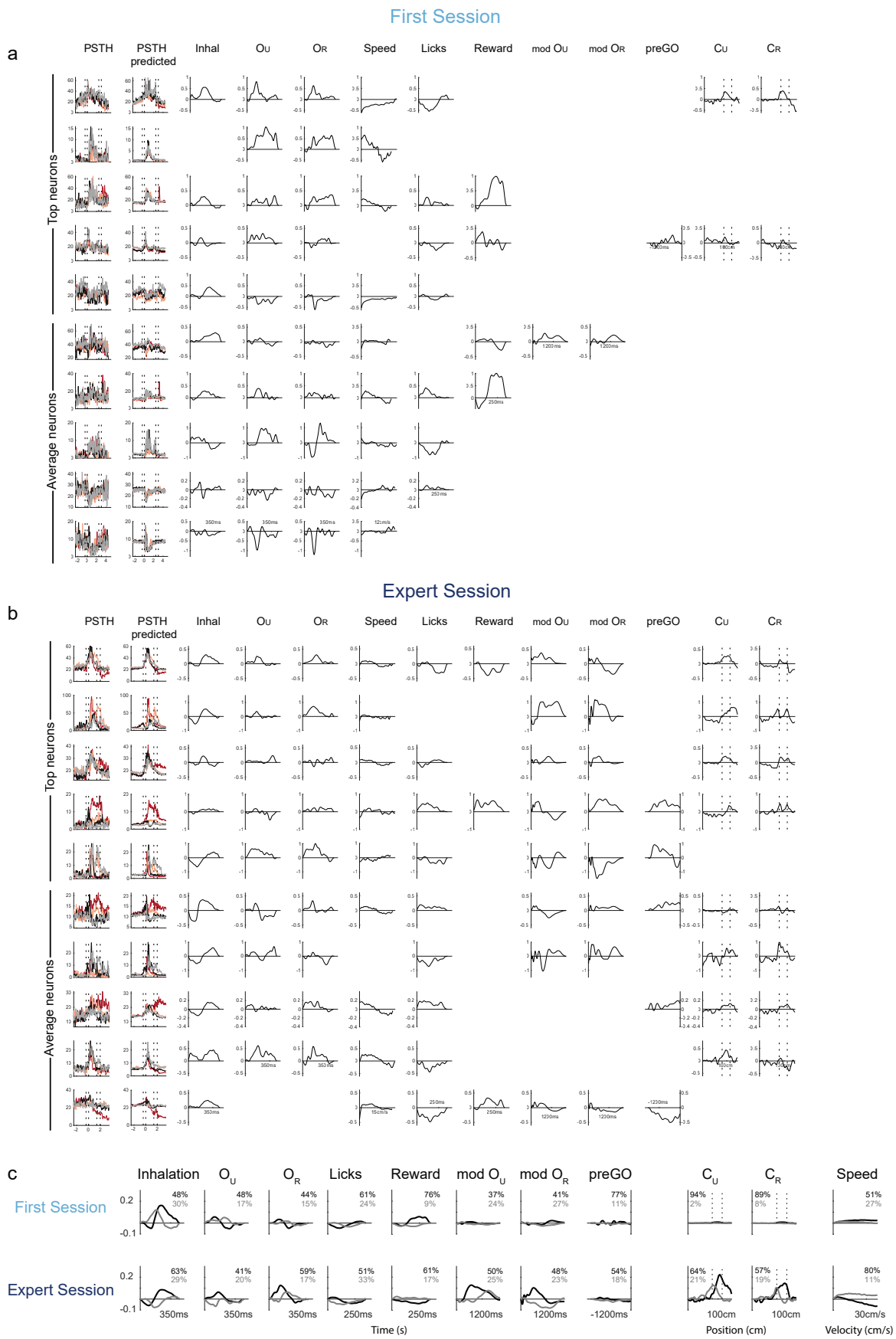
Extended data Fig. 4. Reward anticipation neurons. Differentiating pre-motor and reward anticipatory signals in single neurons. Colors indicate trial outcome: hit and correct rejection in dark and light green, false alarm and miss in dark and light purple. **a**, Behavioral lick response aligned to different trial events. Top panels show raster plot of licks, bottom panel show instantaneous average lick rate. The column on the right shows in gray 29 licks that occurred before odor stimulation and were temporally isolated from other licks by at least 0.75s. **b**, Example of a neuron that is modulated by motor lick signals. Top panels show recorded neuronal activity aligned to different trial events, with spike raster plots on top and instantaneous average neuronal firing rates below. Horizontal dotted lines indicate mean neuronal firing rate across trials. The column on the right shows in gray the instantaneous average neuronal firing rate aligned to isolated licks. Vertical dashed lines indicate a time window between -0.75 s and 0.75 s around lick onset: the neuronal firing rate curve during this time window is the estimated neuronal response to licks, fr^{licks} . Bottom panels show the simulated neuronal activity expected by the lick response. Simulations were obtained by linearly convolving fr^{licks} with the individual lick events across trials (plus a firing rate offset corresponding to the mean neuronal firing rate).

across trials). Notice how simulated activity partially explains the observed instantaneous average neuronal firing rates shown on top, indicating the presence of motor modulations in this neuron. **c**, Example of a neuron with reward anticipation signals that cannot be attributed to motor activity. Notice the complete absence of neuronal firing activity around isolated licks, leading to a flat fr^{licks} that results in a simulated neuronal activity that fails to recapitulate the observed neuronal activity.



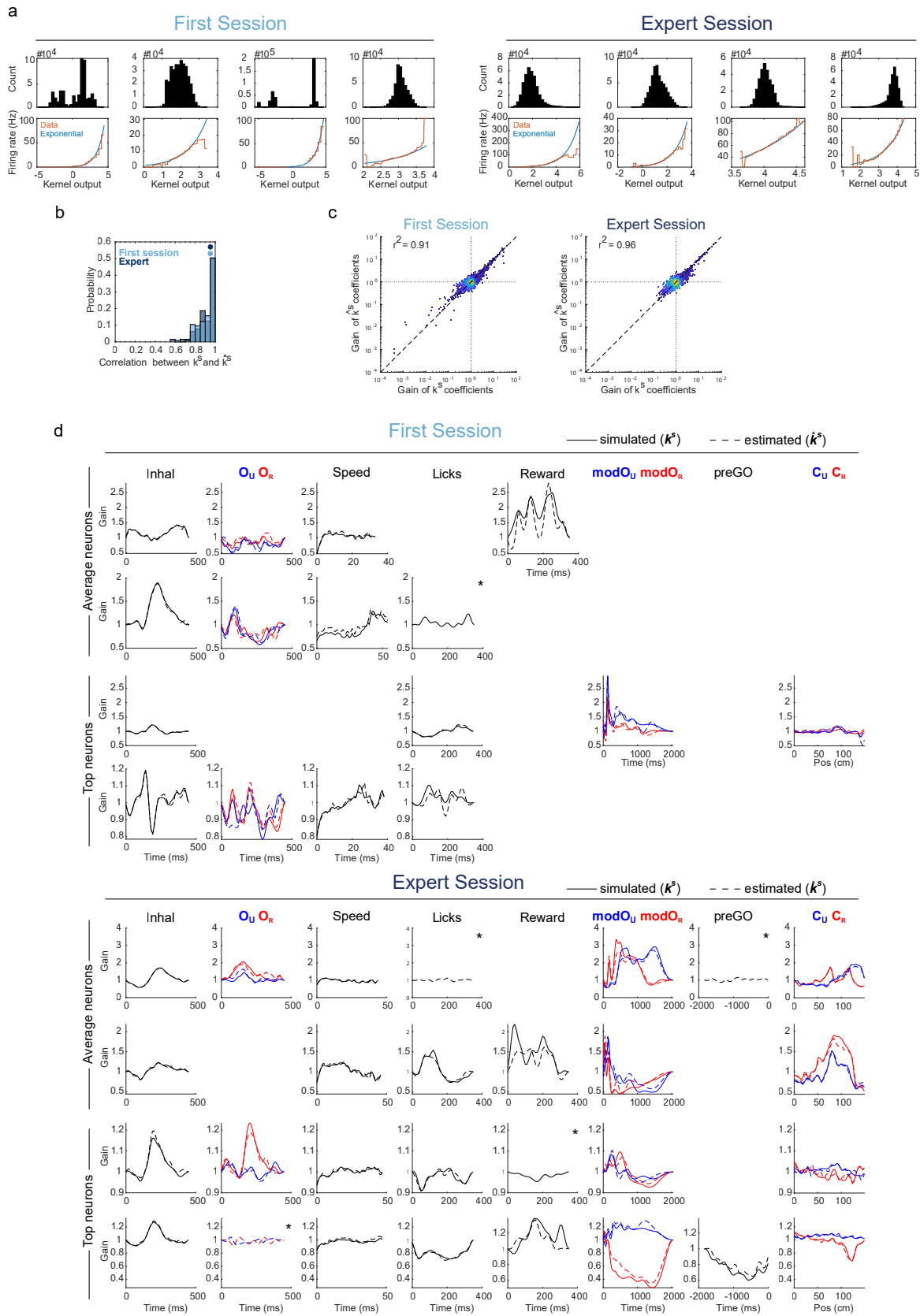
Extended data Fig. 5. Parametrizations in GLM model. a, Scheme of parametrizations of task

variables. Top: time evolution of variables parametrized for convolution with event-based kernels. Bottom: animal-state variables parametrized for convolution with tuning-based kernels. The instantaneous variable values in two individual 10-ms time bins are labelled in purple and cyan. **b**, Scheme of basic functions used to parametrize kernels: an example for an event-based kernel (top panel; inhalation kernel) and an example for a tuning-based kernel (bottom panel; spatial context kernel). **c**, Model-based PSTH and single-trial predictions. Examples of four neurons are shown (two "top neurons" and two "average neurons", according to PSTH prediction accuracy). Each row shows the neuron's measured and predicted PSTHs (first two columns) along with the measured and predicted single trials (last four columns, two "best trials" and two "median trials" according to trial prediction accuracy). For single trials, black ticks at the bottom indicate the observed spike train and the black trace shows an estimate of spike rate obtained by smoothing the spike train with a 300-ms boxcar, while the orange trace shows the model-based spike rate prediction for that trial. **d**, Contribution of each task variable to the GLM encoding model (bits per spike) in first and expert sessions. Black dots indicate statistically significant differences between first or expert session recordings (Wilcoxon rank sum test. I, p-value = 2.3×10^{-2} ; L, p-value = 2.6×10^{-2} ; R, p-value = 3.8×10^{-2} ; M, p-value = 6.9×10^{-8} ; G, p-value = 1.6×10^{-4} ; C, p-value = 8.12×10^{-18}) **e**, Standard deviation of kernels as a function of each variable in first and expert sessions. Asterisks indicate statistically significant differences. Wilcoxon rank sum test was used for first session vs. expert session comparisons (top asterisks: R, p-value = 4.5×10^{-2} ; ModOU, p-value = 2.7×10^{-7} ; ModOR, p-value = 6.1×10^{-8} ; G, p-value = 1.1×10^{-4} ; CU, p-value = 6.4×10^{-18} ; CR, p-value = 5.2×10^{-18}) and Wilcoxon signed rank paired test was used for comparison across kernel categories of each neuron (bottom asterisks: modOU vs modOR, p-value = 2.6×10^{-2}).



Extended data Fig. 6. Examples of kernels obtained with the GLM encoding models of

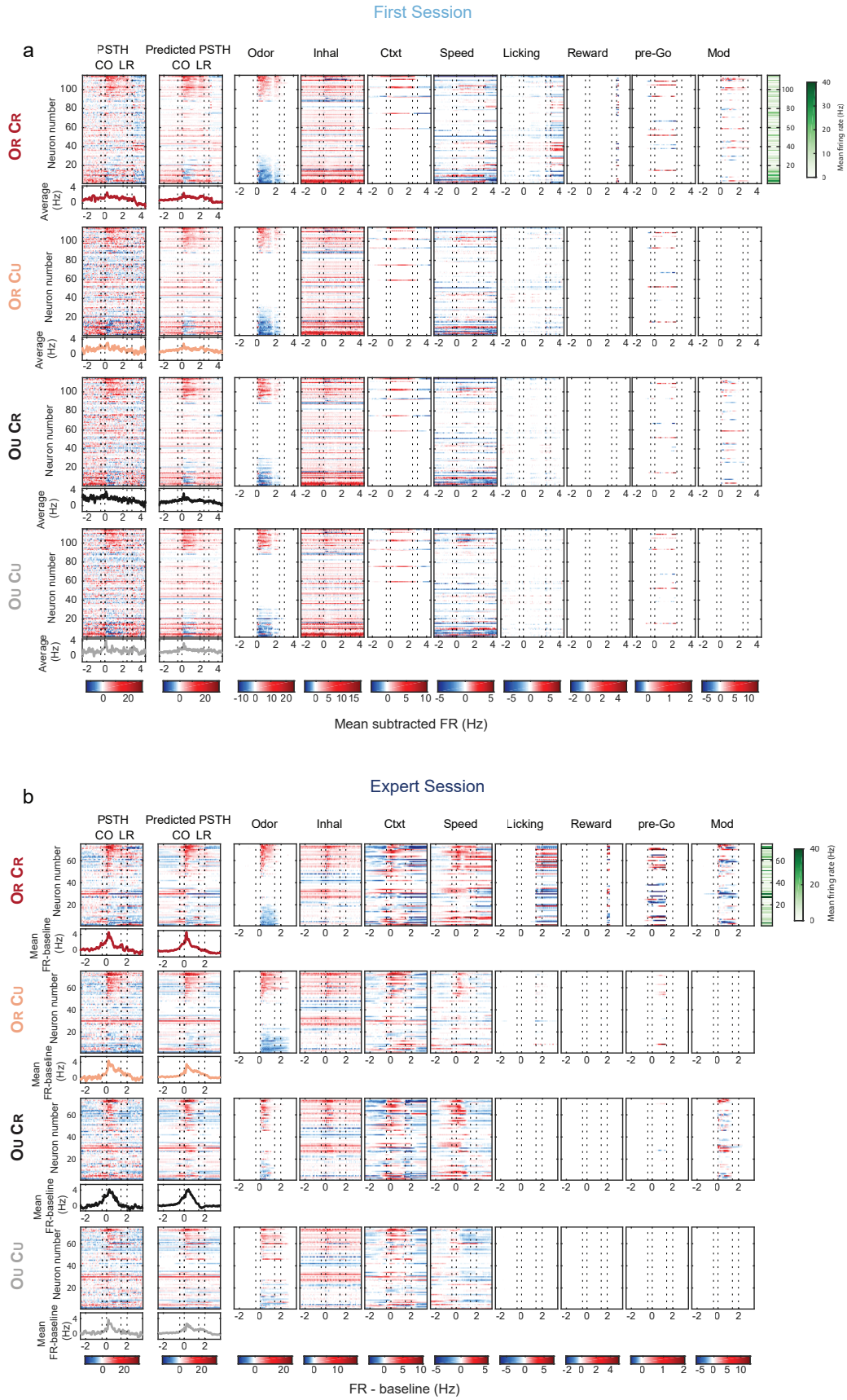
neurons. a and b, Examples of PSTHs and model-predicted PSTHs of several neurons recorded during first session (a) and during expert session (b). Panels on the right show the corresponding kernels obtained for each neuron. **c,** First (black) and second (grey) PCA component of pooled kernels for inhalation, unrewarded odour (O_U), rewarded odour (O_R), licking, reward, modulation of context (mod) onto O_U , mod O_R , activity before a GO decision (preGO), unrewarded context (C_U), rewarded context (C_R) and animal speed. Explained variance by each component is indicated in percentages, first-session recordings (top) and expert-session recordings (bottom). Parametrizations in GLM model.



Extended data Fig. 7. Evaluation of the accuracy of GLM models. a, Validation of GLM

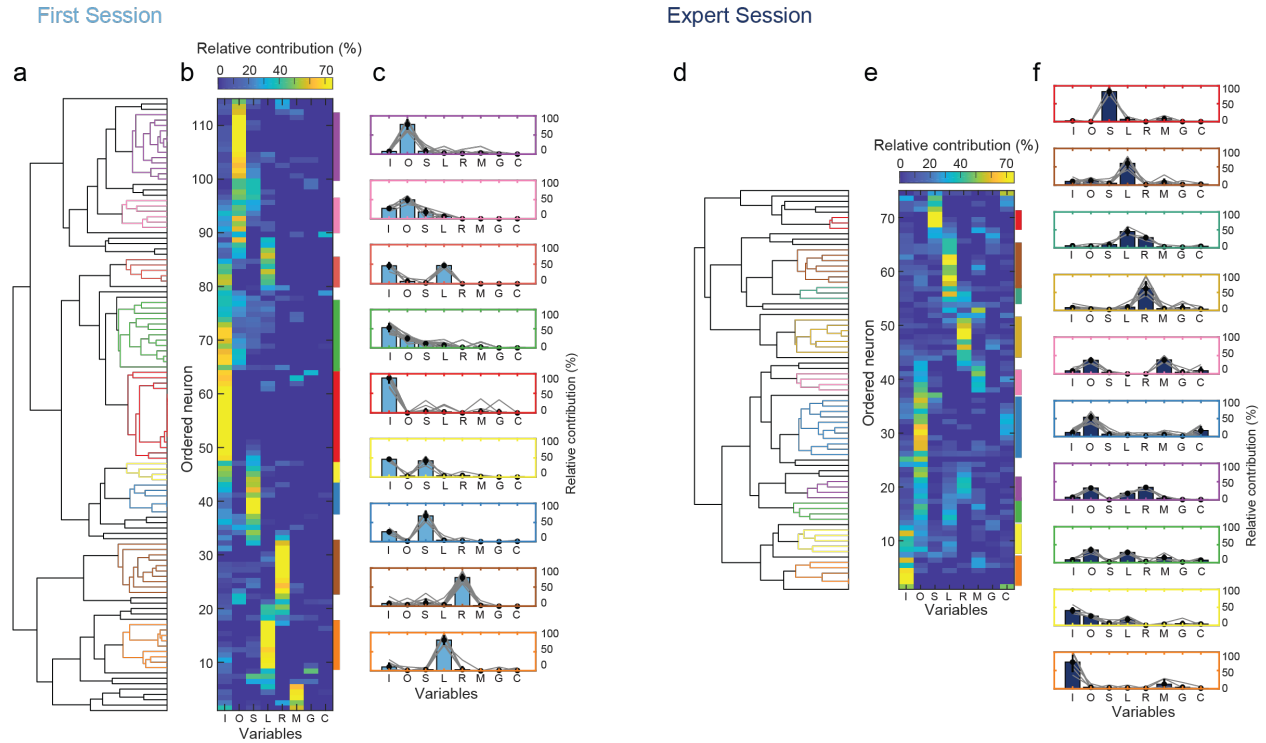
exponential nonlinearity. Data for four typical neurons of first session (left) and expert session (right) recordings are shown. Top panels: histograms of kernel output (i.e., $\mathbf{k} \otimes \mathbf{x}$) values for each neuron. Bottom panels: the observed spike rate for each of the kernel output bins (orange) and the predicted obtained by applying an exponential non-linearity to the kernel output bins (blue).

b-d, Effectiveness of GLM model in quantifying the contributions of variables to the neuronal response (see *Methods*). **b**, Histogram of Pearson correlation coefficients between simulated kernels ($\mathbf{k}^s$) and estimated kernels ($\hat{\mathbf{k}}^s$) for all neurons simulated according to the recorded task data in first session and expert session animals. Dots indicate median correlation values (0.95 and 0.951 for first session and expert session, respectively). **c**, The relation between the multiplicative neuronal gain of individual $\mathbf{k}^s$ and $\hat{\mathbf{k}}^s$ coefficients. Color code indicates density of data. Diagonal dashed line indicates one-to-one relation. Vertical and horizontal dotted lines indicate a gain coefficient of 1 for $\mathbf{k}^s$ and $\hat{\mathbf{k}}^s$. Data for first and expert session are shown on the left and right panels, respectively. Coefficients of determination (r^2) are indicated. **d**, Examples of simulated $\mathbf{k}^s$ (solid lines) and estimated $\hat{\mathbf{k}}^s$ (dashed lines) kernels for average (median correlation between $\mathbf{k}^s$ and $\hat{\mathbf{k}}^s$) and top (highest correlation between $\mathbf{k}^s$ and $\hat{\mathbf{k}}^s$) neurons. Asteriks mark discrepancies in which kernels were not detected (i.e., included in $\mathbf{k}^s$ simulations but missing in $\hat{\mathbf{k}}^s$) or erroneously estimated (i.e., not included in $\mathbf{k}^s$ simulation but estimated in $\hat{\mathbf{k}}^s$), which were typically for kernels of multiplicative gain ~ 1 which have negligible impact on neuronal response encoding.

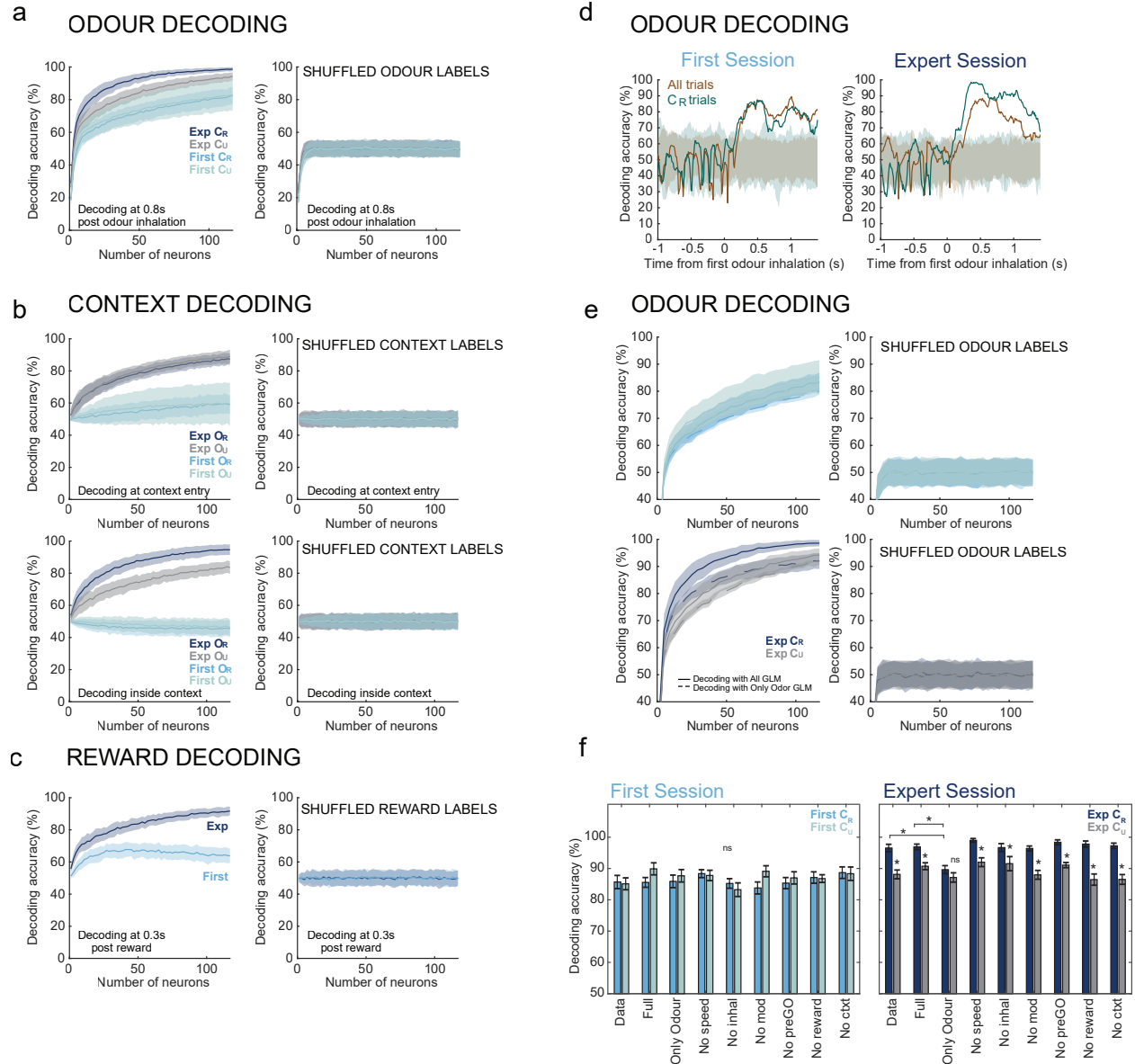


Extended data Fig. 8. Firing-rate modulations induced by individual model variables. a,

PSTHs and model-predicted PSTHs of all neurons with fitted kernels, recorded in first sessions. Panels on the right show predicted firing rate modulations (around the neuron's mean firing rate) induced by each variable (i.e., for GLM models including only that task variable). Neurons were sorted according to their average responses to ORCR trials. **b**, Same as a but for expert sessions.

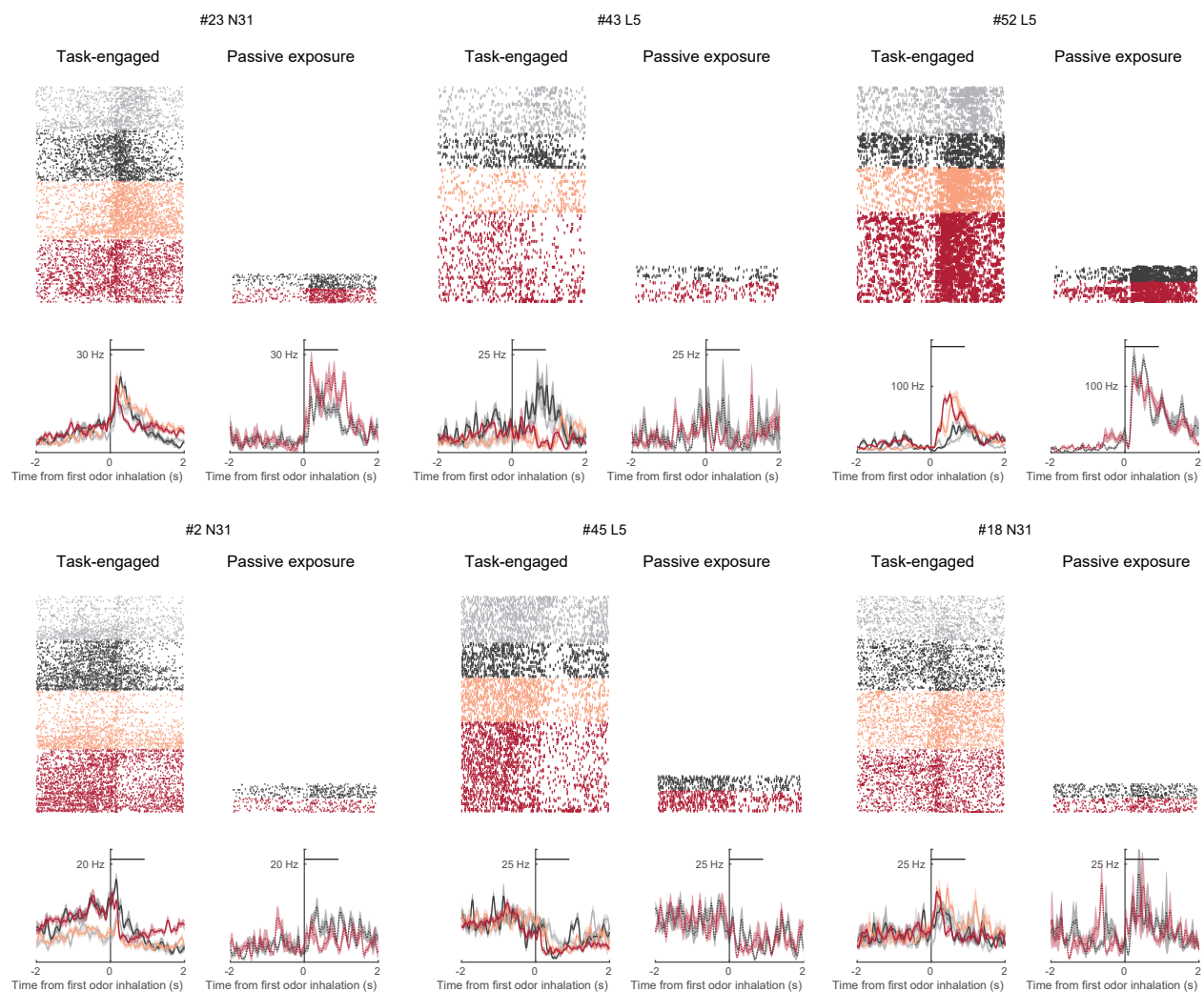


Extended data Fig. 9. Clustering neurons according to the relative contribution of their encoded task variables. **a and d**, Dendrograms for the hierarchical tree of first-session (a) and expert-session (d) neurons sorted according to the relative contribution of their encoded task variables. **b and e**, Relative contributions of task variables to first-session (b) and expert-session (e) neurons, sorted according to the hierarchical tree. Colour bars on the right indicate each individual cluster obtained (same colour code applies to dendrograms). **c and f**, Bar plot for each individual first-session (c) or expert-session (f) cluster, showing the relative contribution of the variables encoded by the neurons in the cluster (same data shown in Fig. 4e).

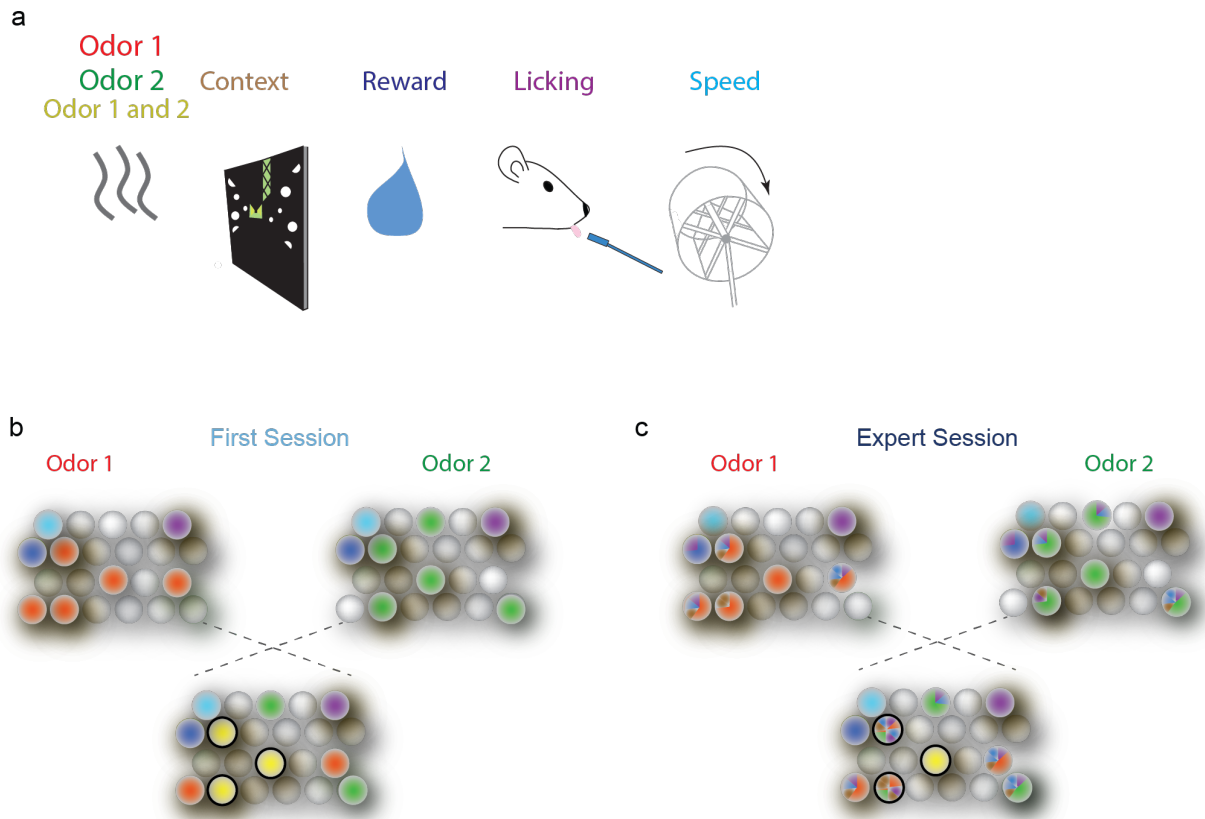


Extended data Fig. 10. Multidimensional decoding in piriform neurons. **a-c**, GLM-based trial-by-trial decoding as a function of neuronal population size, using first-session (First) and expert-session (Exp) recordings, and after shuffling trial labels. **a**, Odour identity decoding at 0.8s after first odorant inhalation in rewarded and unrewarded context trials (C_R and C_U). **b**, Context identity decoding in rewarded and unrewarded odour trials (O_R and O_U). Top, decoding at the moment of context entry. Bottom, decoding when mouse is inside context zone. **c**, Decoding reward consumption 0.3s after reward delivery. **d**, Cross-validated time-dependent odour identity decoding accuracies of a linear classifier given by the leading odour component of dPCA applied to piriform activity, using either all trials (brown line) or C_R trials only (green line). Shaded regions show distribution of decoding accuracies expected by chance as estimated by 100 iterations of a trial shuffling procedure. Statistically significant decoding is obtained

when the accuracy exceeds the accuracy by chance. First-session and expert-session data are shown on the left and right panels, respectively. **e**, Same GLM-based decoding analysis as in **a**, but decoding recorded data using GLM models including all fitted kernels (All GLM) and after removing contributions of non-odour-related kernels from the simulations of odour-responsive neurons (Only Odour GLM). Top, decoding accuracy for first-session data in C_R and C_U trials. Bottom, same as Top but for expert-session. Right panels show results after shuffling odour labels. For all decoding analysis shown in the figure the total number of trials used for training and testing GLM-based decoders was matched across the different trial types compared. **f**, Decoding accuracy of linear classifiers for odour identity from data (Data) and GLM model simulations, in rewarded and unrewarded context trials (C_R and C_U , respectively), for first-session (First) and expert (Exp) animals. Odor identity was decoded with the activity of a population of 100 neurons, during a 0.5s window after first odorant inhalation. Simulations were obtained for GLM models including all fitted kernels (All GLM), GLM models where we removed from odour-responsive neurons all the non-olfactory kernels (Only Odour), and GLM models where we removed from odour-responsive neurons the following single kernels: context (No Ctxt), speed (No Speed), inhalation (No Inhal), modulation of odour responses by rewarded context (No Mod), anticipation of GO response (no PreGO) and reward consumption (No Reward). 2-Way Anova, for context trial type (C_R and C_U ; 1 degree of freedom), Data type (Data and simulations from the 8 GLM model types; 8 degrees of freedom). Context-Data type interaction $\text{Prob} > F: 2 \times 10^{-5}$. One-way ANOVA corrected for multiple comparisons: Data C_R vs Full C_R , n.s.; Data C_U vs Full C_U , n.s.; Full C_R vs Only Odor C_R , $p\text{-value} = 2 \times 10^{-6}$; Full C_U vs Only Odor C_U , n.s.; Full C_R vs every GLM simulation with single kernels removed C_R , n.s.; Full C_U vs every GLM simulation with single kernels removed C_U , n.s.; Data C_R vs Data C_U , $p\text{-value} = 7 \times 10^{-7}$; Full C_R vs Full C_U , $p\text{-value} = 1 \times 10^{-4}$; No Ctxt C_R vs No Ctxt C_U , $p\text{-value} = 7 \times 10^{-7}$; No Speed C_R vs No Speed C_U , $p\text{-value} = 7 \times 10^{-6}$; No Inhal C_R vs No Inhal C_U , $p\text{-value} = 7 \times 10^{-3}$; No Mod C_R vs No Mod C_U , $p\text{-value} = 7 \times 10^{-7}$; No PreGO C_R vs No PreGO C_U , $p\text{-value} = 2 \times 10^{-6}$; No Reward C_R vs No Reward C_U , $p\text{-value} = 7 \times 10^{-7}$; Only Odor C_R vs Only Odour C_U , n.s.



Extended data Fig. 11. Passive vs task-engaged PCx responses. Example of neuronal recordings during task engagement or when the virtual reality was turn off and animals were passively exposed to odours.



Extended data Fig. 12. Learning induces mixed-selectivity in PCx neurons that improves discrimination. **a**, Variables associated with the odour-visual context-reward associative task. The variables are colour coded. **b**, The scheme represents responses of neurons to different variables from **a**, in an animal in the first session of training. Neurons tend to respond to single variables and do not show contextual modulation. Responses to individual odours have a percentage of overlapping illustrated by the yellow neurons that do not discriminate. **c**, The scheme represents responses of neurons to different variables from **a**, in an animal in the expert session of training. Neurons tend to have associative responses, with a contextual modulation. Notice that the overlap of the responses to the two odours decreases due to the mixed-selectivity that arises with learning the task.